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Research Article

The positive feedback regulatory loop of miR160-Auxin Response Factor 17-HYPONASTIC LEAVES 1 mediates drought tolerance in apple trees

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Abstract

Drought stress tolerance is a complex trait regulated by multiple factors. Here, we demonstrate that the miRNA160–Auxin Response Factor 17 (ARF17)–HYPONASTIC LEAVES 1 module is crucial for apple (Malus *domestica*) drought tolerance. Using stable transgenic plants, we found that drought tolerance was improved by higher levels of Mdm-miR160 or *MdHYL1* and by decreased levels of *MdARF17*, whereas reductions in *MdHYL1* or increases in *MdARF17* led to greater drought sensitivity. Further study revealed that modulation of drought tolerance was achieved through regulation of drought-responsive miRNA levels by MdARF17 and MdHYL1; MdARF17 interacted with MdHYL1 and bound to the promoter of *MdHYL1*. Genetic analysis further suggested that *MdHYL1* is a direct downstream target of MdARF17. Importantly, MdARF17 and MdHYL1 regulated the abundance of Mdm-miR160. In addition, the Mdm-miR160-MdARF17–MdHYL1 module regulated adventitious root development. We also found that Mdm-miR160 can move from the scion to the rootstock in apple and tomato (*Solanum lycopersicum*), thereby improving root development and drought tolerance of the rootstock. Our study revealed the mechanisms by which the positive feedback loop of Mdm-miR160-MdARF17-MdHYL1 influences apple drought tolerance.

Introduction

MicroRNAs (miRNAs) are endogenous noncoding regulatory RNAs transcribed by RNA polymerase II to produce primary miRNAs (pri-miRNAs). In plants, DICER-LIKE 1 (DCL1) processes most pri-miRNAs by cleavage. Subsequently, the interacting proteins SERRATE (SE), HYPONASTIC LEAVES 1 (HYL1), and nuclear cap-binding complex process the pri-miRNAs to form short duplex miRNAs, which consist of a mature miRNA and a passenger miRNA strand (Akdogan et al., 2016; Aravind et al., 2017; Yang et al., 2019), ∼20–24 nucleotides in length (Eldem et al., 2013; Alptekin et al.,

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2017; Swapna and Kumar, 2017). Studies have shown that mature miRNAs bind to the target mRNA through the formation of an RNA-induced silencing complex, thereby behaving as negative regulators of target genes (Samad et al., 2017; Swapna and Kumar, 2017).

Plant miRNAs play crucial roles in abiotic stress responses (Ma et al., 2015; Song et al., 2019). Overexpression (OE) of Arabidopsis (*Arabidopsis thaliana*) ath-miR156, soybean (*Glycine max*) gma-miR172c, rice (*Oryza sativa*) osa-miR319, Solanaceae miR396a-5p, and ath-miR408 improves tolerance to drought stress in various plant species (Zhou and Luo, 2014; Chen et al., 2015a, 2015b; Hajyzadeh et al., 2015; Li et al., 2016; Arshad et al., 2017). Knocking down osa-miR166 leads to improved drought tolerance through the regulation of stem xylem development (Zhang et al., 2018a, 2018b). Different members of a single miRNA family may play different roles in plant stress tolerance. Elevated expression of tomato (*Solanum lycopersicum*) sly-miR169c reduces stomatal conductance and transpiration rate (Zhang et al., 2011), whereas increased expression of ath-miR169a enhances sensitivity to drought stress in Arabidopsis (Li et al., 2008). In addition, higher levels of osa-miR393 reduce the drought tolerance of rice, but osa-miR393a increases the drought tolerance of creeping bentgrass (*Agrostis stolonifera* L.) (Xia et al., 2012; Zhao et al., 2019). In the apple (*Malus domestica*) genome, 23 conserved, 10 less-conserved, and 42 apple-specific miRNAs or families were initially identified (Xia et al., 2012). Subsequently, using an F_1 hybrid population from a cross between a drought-tolerant genotype and a drought-sensitive genotype, 67 miRNAs that may play critical roles in apple drought stress tolerance were recognized (Niu et al., 2019). For example, apple mdm-miR156 and mdm-miRn249 are two positive regulators of apple osmotic stress tolerance (Niu et al., 2019).

Regulation of miRNA genes (MIR) is facilitated by transcription factors (TFs). Two SQUAMOSA PROMOTER BINDING PROTEIN-LIKE (SPL) genes, *SPL9* and *SPL10*, which are repressed by miR156, promote the transcription of miR172b (Wu et al., 2009). POWERDRESS promotes the transcription of MIR172a, b, and c, and enhances Poll II enrichment on their promoters (Yumul et al., 2013). Classes II and III homeodomain leucine zipper proteins act together to repress transcription of MIR165/166 through a conserved cis-element on their promoters (Merelo et al., 2016). Two Heat Shock TFs, HSFA1b and HSFA7b, can recognize the heat shock element (HSE) element on the promoter of miR398b and activate expression of miR398b, thus improving plant thermotolerance (Guan et al., 2013a, 2013b). Moreover, some TFs are targeted by their cognate miRNAs, thereby regulating MIR expression in a complex model of regulatory loops. The study demonstrated that APETALA2 (AP2) binds to two MIRs that affect AP2 expression, with AP2 positively regulating miR156 and negatively regulating miR172 (Iniguez et al., 2015). A regulatory loop is also involved in HSFA1b/HSFA7b-mediated miR398b expression in response to heat stress (Guan et al., 2013a, 2013b).

In Arabidopsis, miR160-targeted AUXIN RESPONSE FACTORS (ARF10, ARF16, and ARF17) regulate several aspects of plant development (Wang et al., 2005). Disruption of miR160 regulation of ARF17 results in developmental defects of vegetative and floral organs, including roots. *5mARF17* (a resistant version of miR160-directed cleavage) plants, in which ARF17 over-accumulates, display reduced primary root length and lateral root numbers (Mallory et al., 2005). In addition, ARF17 plays critical roles in adventitious root initiation and pollen wall pattern formation in Arabidopsis (Gutierrez et al., 2009; Yang et al., 2013). However, the importance of miR160-mediated ARF17 in drought response has not been explored, particularly in woody plants.

Water shortage is one of the most important abiotic stresses that limit crop growth and development, and has caused total crop losses of approximately $30 billion in recent decades (Gupta et al., 2020). With population growth and the acceleration of global warming, the demand for drought-tolerance breeding is increasing. Therefore, it is important to understand how crops—including fruit crops—respond to drought stress. Plants have evolved several physiological and cellular strategies to cope with drought stress, including hormone content, metabolites, root system architecture, membrane integrity, and antioxidant enzyme activity (Nadarajah and Kumar, 2019). The plant root system is the major organ for absorbing water and nutrients from the soil. To avoid the drought stress, plants would modify its roots to grow deeper into the soil (Kim et al., 2020). For example, higher expression of DEEP ROOTING 1 in rice promotes deeper root growth, resulting in better performance under drought (Uga et al., 2013). In addition, WUSCHEL-related homeobox gene PagWOX11/12a can promote root elongation and biomass, thereby increasing drought tolerance of poplar (*Populus alba* × *Populus glandulosa*; Wang et al., 2020). At the molecular level, many drought-responsive genes have been identified, including both regulatory and functional genes (Govind et al., 2009; Harb et al., 2010; Thirunavukkarasu et al., 2017). TFs are regulatory proteins that directly or indirectly modulate the expression of a specific set of genes. A single TF can modulate the expression of a number of genes. Therefore, it is important to identify key TFs and the networks by which they regulate drought tolerance, especially miRNA-targeted TFs and their targets.

Grafting has been widely used to facilitate flowering, dwarfing, and stress tolerance in fruit crops (Wang et al., 2019). Grafting can trigger the communications of mRNAs, small RNAs, proteins, and phytohormones between scions and rootstocks (Nanda and Melnyk, 2018; Pagliarani and Gambino, 2019). Several studies have identified the transmissible mRNAs and proteins involved in plant development, phytohormone regulation, and stress response (Kasai et al., 2011; Xu et al., 2017; Liu and Chen, 2018; Wang et al., 2019; Hao et al., 2020). Depending on their size, sequence, stability, or secondary structure, miRNAs can move from cell to cell or over long distances (Melnyk et al., 2011; Pagliarani and Gambino, 2019). Mobile miRNAs are involved in some physiological processes and development. For instance, miR399 and miR827 act as long-distance signaling molecules between scions and rootstocks to regulate plant phosphate homeostasis (Pant et al., 2008; Huen et al., 2017); miR172 can move long distances in tobacco and this movement from scions to rootstocks can induce tuberization in potato (Martin et al., 2009; Kasai et al., 2011). Grafting transmissible miR156 modulates plant architecture and tuberization in potato (Bhogale et al., 2014). Mobile small miRNA165/166 regulate the cell differentiation in a gradient of dose-dependent manner in Arabidopsis (Furuta et al., 2012). miR395 can be transmitted through graft unions from scions to rootstocks, leading to the downregulation of its target, *APS4* (Ai et al., 2016). However, so far, roles of transmissible miRNAs in plant drought response are not clear.

Apple trees are often damaged by frequent drought stress, which reduces both fruit production and fruit quality (Li et al., 2019; Niu et al., 2019). However, the role of miRNA-targeted TFs and their targets in apple drought stress is poorly understood. Here, we explore the role of Mdm-miR160, its target *MdARF17*, and MdHYL1 in apple drought response. We demonstrate that the positive feedback loop of Mdm-miR160-MdARF17-MdHYL1 is critical for apple survival under drought stress.

Results

Apple ARF17 is repressed by drought stress in *Malus sieversii* and *M. × domestica*

Previously, we found that expression of *MsARF17* from *M. sieversii* was reduced under simulated drought conditions (Figure 1A; Geng et al., 2018). In *M. × domestica* cv *Golden Delicious*, *MdARF17* was also repressed by drought stress in a time-dependent manner (Figure1B). Tissue-specific expression assays demonstrated that *MdARF17* was expressed in roots, stems, and leaves, with its greatest abundance in flowers (Figure 1C). Protein localization analysis suggested that MdARF17 was localized in the nucleus (Figure 1D). The apple (*M. × domestica*) genome contains one paralog of MdARF17, MdARF17-like. We found that the expression pattern in response to drought, tissue-specific expression, and subcellular localization of MdARF17-like were similar to those of MdARF17 (Supplemental Figure S1), indicating that MdARF17-like may have a function similar to that of MdARF17 in response to drought.

MdARF17 negatively regulates drought tolerance and adventitious root development

We next generated *MdARF17* transgenic plants. Because of the high sequence similarity between *MdARF17* and *MdARF17*-like, we knocked down both genes simultaneously using an RNAi approach (Supplemental Figure S2, A and B). To determine whether the RNAi approach knocked down other *MdARF* genes, we examined the expression of additional *MdARF* genes that were highly similar to *MdARF17*: *MdARF3*, *MdARF18*, and *MdARF18-like*. We found that expression of these *MdARF* genes was not affected (Supplemental Figure S2, C–E). A mutant version of *MdARF17* (*MdmARF17*) driven by the 35S promoter was used to generate OE plants of *MdARF17* (Supplemental Figure S3).

To characterize the biological functions of MdARF17 in response to drought stress, we exposed nontransgenic (GL-3) plants and transgenic plants to a short-term drought stress. After short-term drought stress, ∼80% of the *MdARF17* RNAi plants survived compared with only 20% of the GL-3 plants (Figure 2, A and B). After drought treatment, the ion leakage rate of the *MdARF17* RNAi plants was consistently lower than that of GL-3 plants (Figure 2C). These data suggest that the *MdARF17* RNAi plants were more tolerant to drought stress. However, after short-term drought stress, *MdmARF17* OE plants had a lower survival rate (∼34%) and a higher rate of ion leakage (Figure 2, D–F). Taken together, our results suggest that MdARF17 is a negative regulator of apple drought tolerance.

It has been reported that ARF17 in Arabidopsis negatively regulates adventitious root development (Kim et al., 2015). Similarly, we found that MdARF17 is also a negative regulator of apple adventitious root development, as demonstrated by longer root length and greater adventitious root numbers in *MdARF17* RNAi plants and shorter root length and fewer adventitious root numbers in *MdmARF17* OE plants (Supplemental Figure S4, A–F). In addition, after long-term drought stress, *MdARF17* RNAi plants had greater root dry weight and root-to-shoot ratio than GL-3 plants, while *MdmARF17* OE plants had less root dry weight and root-to-shoot ratio than GL-3 plants (Supplemental Figure S5, A–D).

MdARF17 regulates expression of drought-responsive genes

To understand the molecular basis for the effect of MdARF17 on apple drought tolerance, we performed genome-wide expression profiling of GL-3 and *MdARF17* RNAi plants using RNA-sequencing. Under control conditions, 1,136 genes were upregulated and 561 genes were downregulated in the *MdARF17* RNAi plants relative to the GL-3 plants (Supplemental Datasets S1 and S2). Under drought conditions, MdARF17 negatively regulated 479 genes and positively regulated 329 genes (Figure 3, A and B; Supplemental Datasets S3 and S4). Among the 479 genes, 208 were induced by drought stress (Figure 3, A and C), whereas 77 genes of the 329 genes were repressed by drought stress (Figure 3, B and D). We selected 10 genes for expression verification by reverse transcription-quantitative polymerase chain reaction (RT-qPCR), and the expression of eight could be verified. These included higher expression of WRKY DNA-binding protein 71 (*MdWRKY71*), Aquaporin TIP1-3-like (*MdTIP1-3like*), Lipid-transfer protein 3 (*MdLTP3*), Expansin A4 (*MdEXP4*), 1-aminocyclopropane-1-carboxylate synthase-like (*MdACS*), Heat shock protein 70 (*MdHSP70*),

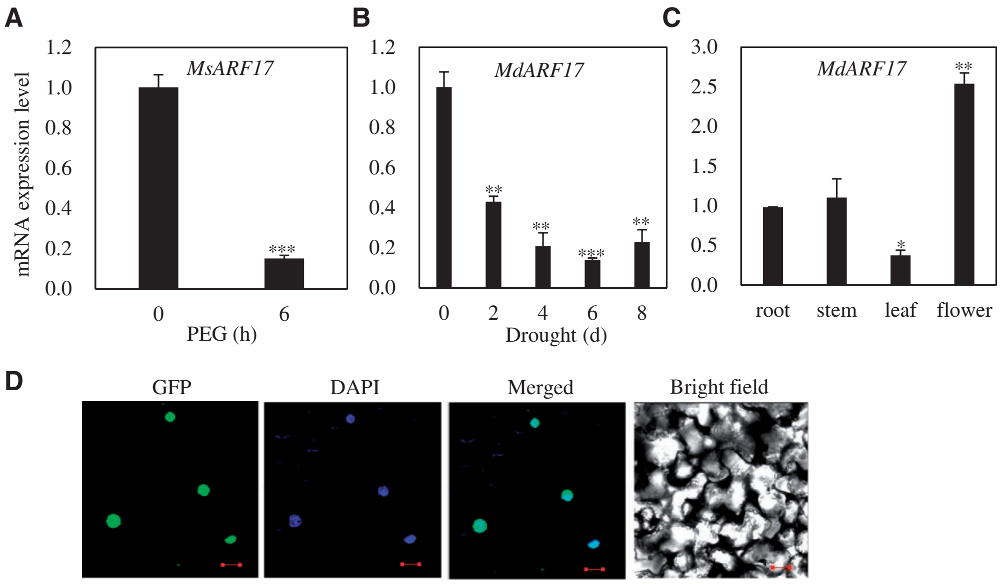

Figure 1 Expression pattern and subcellular localization of MdARF17. A, Expression of *MsARF17* in response to PEG (−0.7 MPa) treatment. *Malus sieversii* plants were hydroponically cultured and treated with PEG8000 for 0 or 6 h, followed by harvesting of roots for RNA extraction. B, Expression of *MdARF17* in response to drought stress. Water was withheld from 2-month-old *M.* × *domestica* for 2, 4, 6, and 8 d, and leaves were collected for RNA extraction. C, Expression of *MdARF17* in different tissues of *M.* × *domestica*. D, Subcellular localization of MdARF17 in epidermal cells of tobacco. Bars = 50 µm. Error bars indicate standard deviation (SD) (n = 3). One-way analysis of variance (ANOVA) (Tukey's test) was performed and statistically significant differences were indicated by *$P < 0.05$; **$P < 0.01$; ***$P < 0.001$.

MADS-box protein(*MdMADS*), and MdSFBB9-alpha gene (*MdSFBB9*) in the *MdARF17* RNAi plants under drought (Figure 3E). Homologs of these confirmed genes are positive regulators of drought stress tolerance (Young et al., 2004; Cho and Hong, 2006; Cho and Choi, 2009; Guo et al., 2013; Khong et al., 2015; Huang et al., 2019; Kurowska et al., 2019).

MdARF17 is a target of Mdm-miR160, a positive regulator of drought tolerance and adventitious root development

Arabidopsis *ARF17* is one of the targets of miR160 (Mallory et al., 2005). Using RNA Ligase-Mediated Rapid Amplification of cDNA Ends (5′-RLM-RACE), we confirmed that apple ARF17 and ARF17-like are two targets of Mdm-miR160 (Figure 4A). We next asked whether Mdm-miR160 plays a role in root development and drought tolerance. We examined expression of Mdm-miR160 in response to drought and found that it was induced by drought stress in a time-dependent manner that was opposite to the expression pattern of *MdARF17* and *MdARF17*-like under drought (Figure 4B). There are five precursor RNAs of Mdm-miR160 in the apple genome. We found that among all the precursors, only primary Mdm-miR160e had an expression pattern similar to that of Mdm-miR160 (Figure 4C; Supplemental Figure S6), indicating that Mdm-miR160 may directly target *MdARF17* and *MdARF17-like* under drought. To verify this targeting under drought in planta, we produced transgenic apple plants that overexpressed Mdm-pri-miR160e (Supplemental Figure S7). We named these transgenic plants Mdm-miR160e OE plants. Under control and drought conditions, expression of *MdARF17* and *MdARF17-like* was repressed in Mdm-miR160e OE plants (Figure 4, D and E), further suggesting that Mdm-miR160 negatively targets *MdARF17* and *MdARF17-like*.

We then examined the drought tolerance of the Mdm-miR160e OE plants. After short-term drought treatment, only ∼20% of the GL-3 plants had survived, but ∼90% of the transgenic plants were still alive (Figure 5, A and B). Consistently, the transgenic plants had a lower level of ion leakage than the GL-3 plants after short-term drought stress (Figure 5C). These results suggest that Mdm-miR160 is a positive regulator of apple drought tolerance.

We also investigated the influence of Mdm-miR160 transcripts on adventitious root development and found that Mdm-miR160e OE plants had greater adventitious root length and root number (Supplemental Figure S8, A–C). In addition, root dry weight and root-to-shoot ratio of Mdm-miR160e OE plants were greater than that of GL-3 after long-term drought stress (Supplemental Figure S8, D and E). These results suggest that Mdm-miR160 is a positive regulator of root development and apple drought tolerance.

MdARF17 interacts with MdHYL1 and negatively regulates *MdHYL1* through directly binding to its promoter

To understand the molecular mechanism by which MdARF17 influences root development and drought stress tolerance, we performed a yeast-two hybrid (Y2H) screen assay. MdARF17 contains two domains: B3 (107–240 amino acid (aa)) and Aux (252–367 aa). Because full-length MdARF17, but not truncated MdARF17 containing B3 or Aux domain, had self-activation activity (Supplemental Figure S9, A and B), we used truncated MdARF17 containing B3 domain as the bait for the

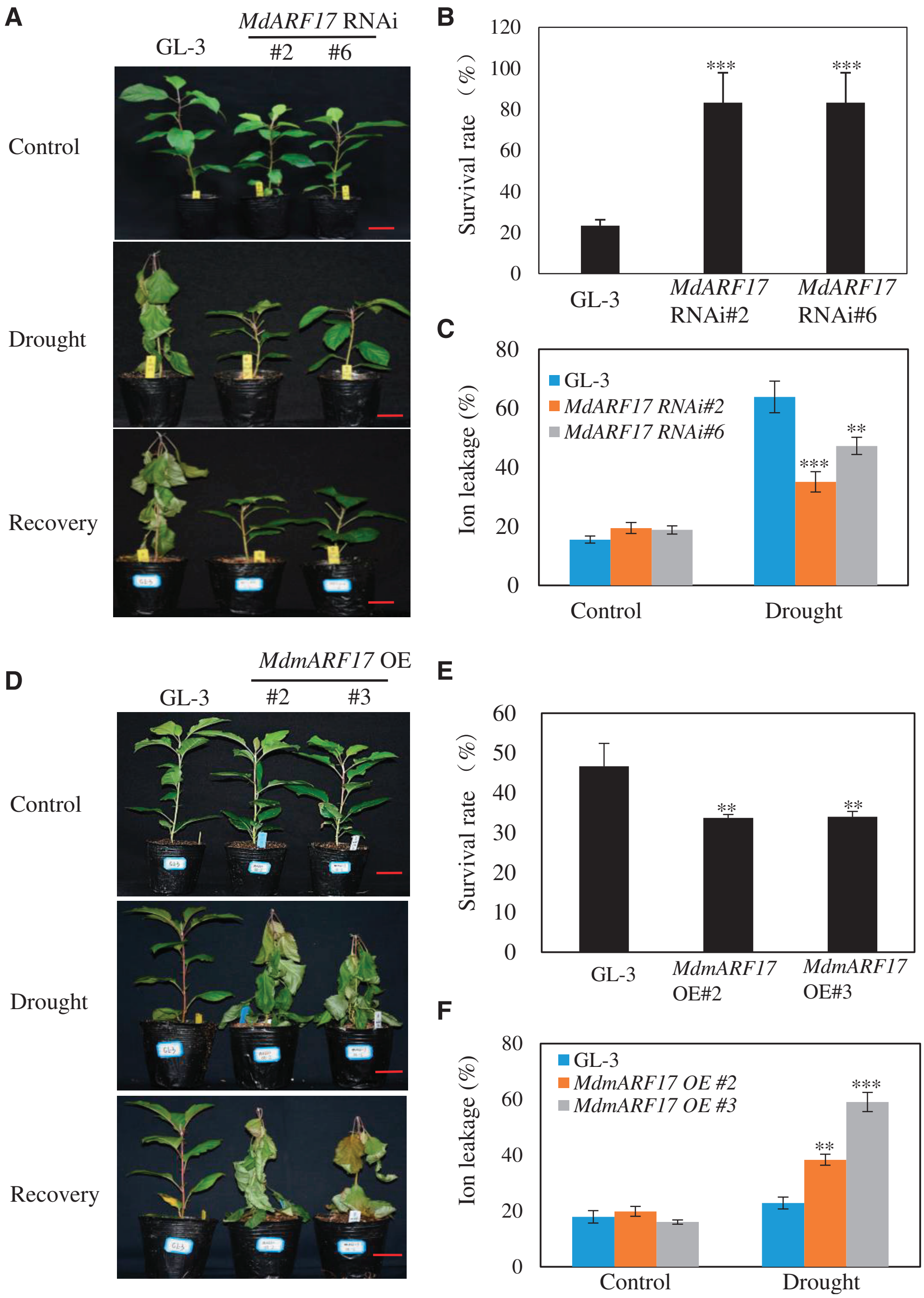

Figure 2 *MdARF17* plays a negative role in apple drought tolerance. A, Drought tolerance of *MdARF17* RNAi plants. Two-month-old plants were transferred to soil for additional 1 month, and water was then withheld for 16 d until VWC decreased to 0. During the treatment, GL-3 plants were supplemented with water to maintain the same VWC with transgenic plants. Bars = 3 cm. B, Survival rate of plants shown in (A). C, Electrolyte leakage of *MdARF17* RNAi plants under control and short-term drought conditions. Plants were exposed to drought until VWC decreased to 5%, and leaves were collected for an ion leakage assay. D, Drought tolerance of *MdmARF17* OE plants. Two-month-old plants were transferred to soil for additional 1 month, and water was then withheld for 14 d until VWC decreased to 0. During the treatment, GL-3 plants were supplemented with water to maintain the same VWC with transgenic plants. Bars = 3 cm. E, Survival rate of plants shown in (D). F, Electrolyte leakage of *MdmARF17* OE plants under control and drought conditions. Plants were exposed to drought until VWC decreased to 5%, and leaves were collected for an ion leakage assay. Error bars indicate SD ($n = 15$ in (B, E)), 15 plants were used and every 5 plants was a biological replication. One-way ANOVA (Tukey's test) was performed and statistically significant differences were indicated by $^{**}P < 0.01$; $^{***}P < 0.001$.

Y2H screen. Among the proteins we identified, one was MdHYL1 which encodes a double-strand RNA binding protein. HYL1 in Arabidopsis is required for accurate processing of miRNAs. AtHYL1 shares 33.13% sequence similarity with MdHYL1, which also contains two double-strand RNA-binding domains (Supplemental Figure S9C). By Y2H, we found

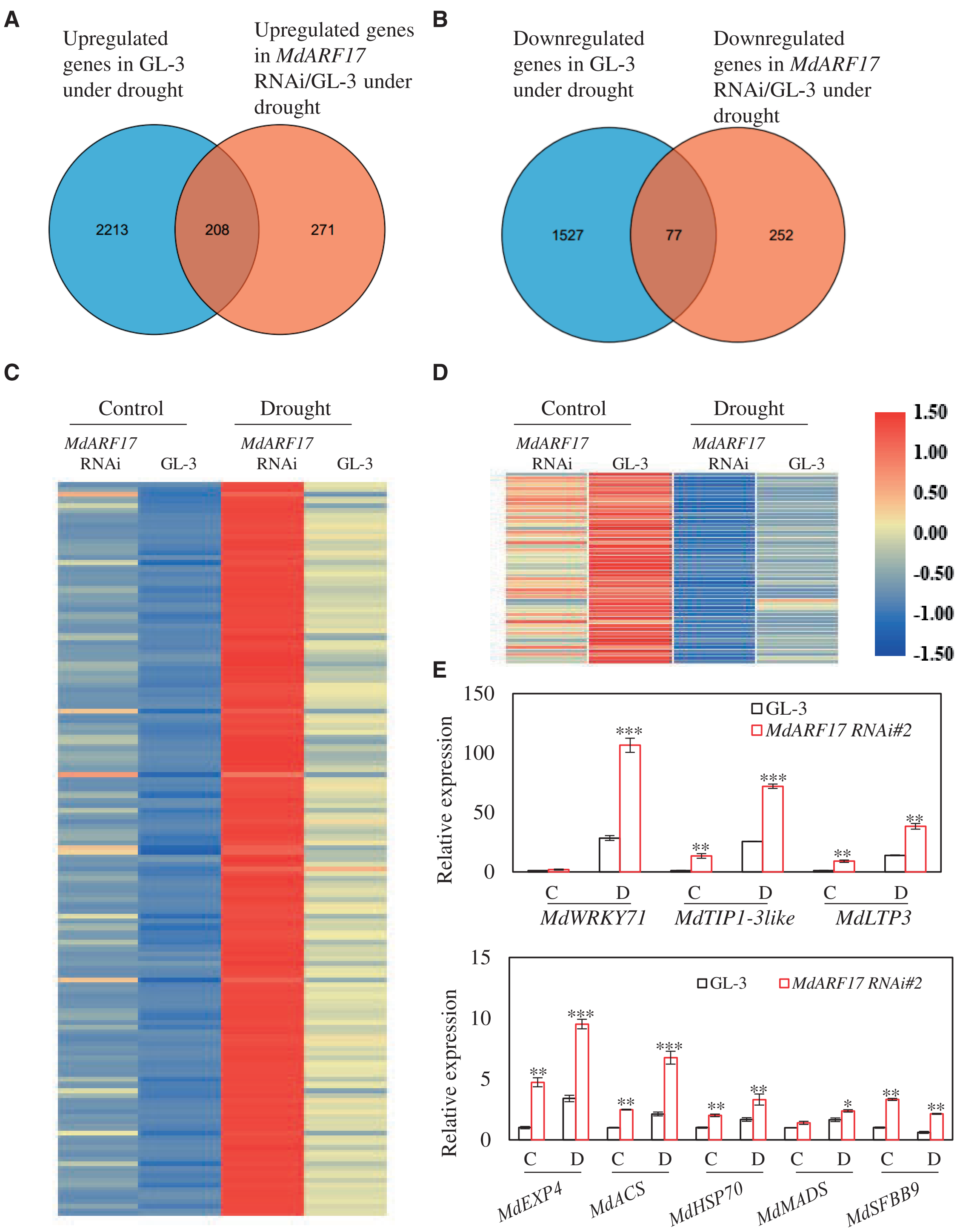

Figure 3 *MdARF17* regulates drought-responsive genes. A, Venn diagrams showing the numbers of genes induced by drought and genes upregulated in *MdARF17* RNAi plants under drought. B, Venn diagrams showing the numbers of genes repressed by drought and genes downregulated in *MdARF17* RNAi plants under drought. C, Heat map showing the $\log_2$(fold change) in transcript level of 208 overlapped genes shown in (A). D, Heat map of 77 overlapped genes shown in (B). E, Verification of transcriptome analysis. Error bars indicate SD (n = 3), three biological replicates were used, and each replicate has three 2-month-old plants. One-way ANOVA (Tukey's test) was performed for data in (E), and statistically significant differences were indicated by $^{*}P < 0.05$; $^{**}P < 0.01$; $^{***} P < 0.001$.

that truncated MdARF17 containing either B3 domain or Aux domain could interact with full-length MdHYL1 (Figure 6A). In addition, truncated MdHYL1 that only contained the two double-strand RNA binding motifs could interact with full-length MdARF17 (Figure 6B; Supplemental Figure S9D), whereas full-length MdHYL1 had self-activation activity (Supplemental Figure S9E). We further confirmed the protein interaction with bimolecular fluorescence

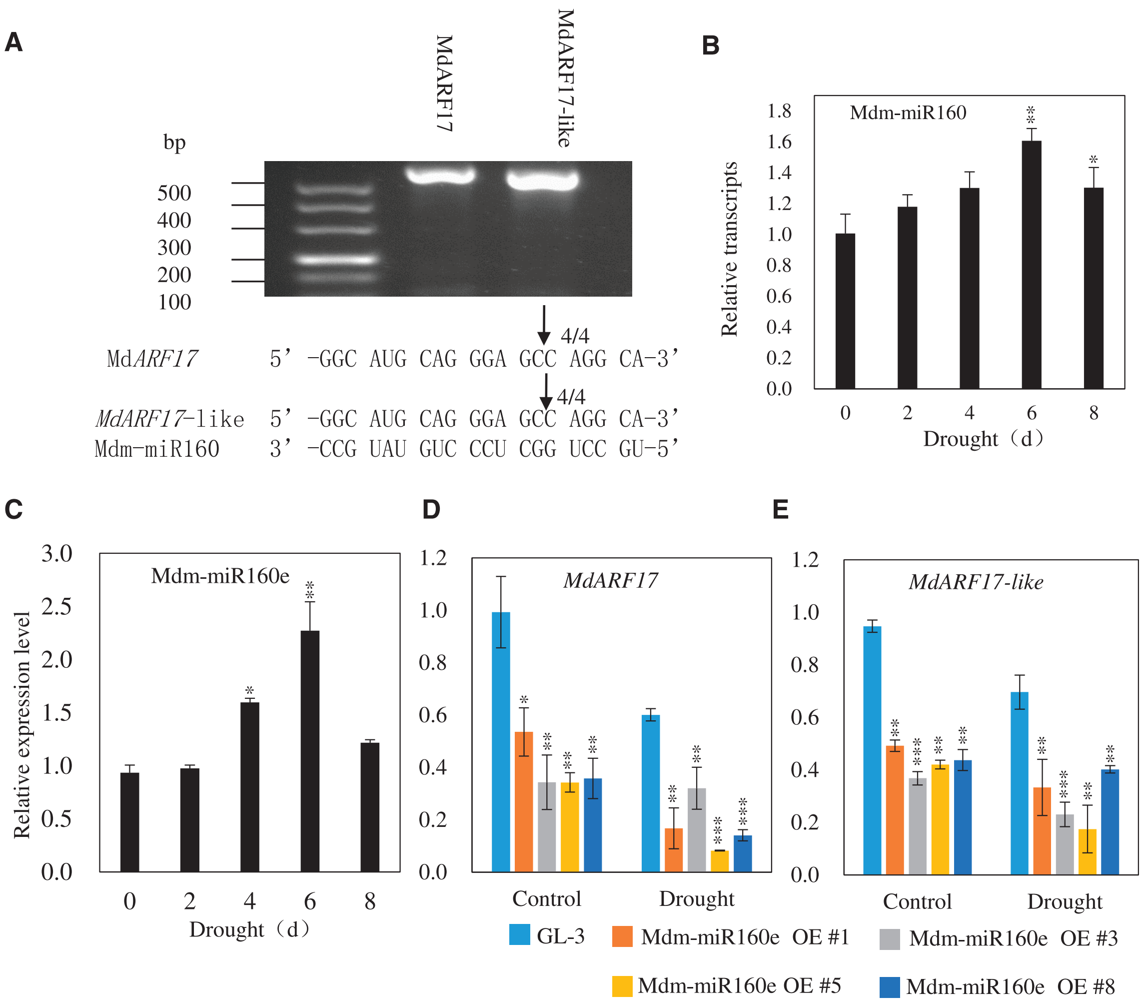

Figure 4 *MdARF17* and *MdARF17*-like are targets of Mdm-miR160 under control and drought conditions. A, 5′-RLM-RACE analysis. B and C, Mdm-miR160 and Mdm-miR160e expression in response to drought. *Malus domestica* grafted on *M. hupehensis* was withheld from water for 0, 2, 4, 6, 8 d, and leaves were collected for RNA extraction. D and E, Expression of *MdARF17* and *MdARF17*-like in Mdm-miR160e OE plants under control and drought conditions. Error bars indicate SD (n = 3 in (B–E)), three biological replicates were used, and each replicate has three 2-month-old plants. One-way ANOVA (Tukey's test) was performed and statistically significant differences were indicated by $^*P < 0.05$; $^{**}P < 0.01$; $^{***}P < 0.001$.

complementation (BiFC) and co-immunoprecipitation (Co-IP) assays (Figure 6, C and D). Subcellular localization analysis suggested that MdHYL1 was co-localized with MdARF17 in the nucleus (Supplemental Figure S10, A and B). In addition, *MdHYL1* was expressed in almost all tissues (Supplemental Figure S10C).

In Arabidopsis, ARF17 often negatively regulates the expression of its target genes by associating with the TGTCTC cis-element in the target promoters (Hayashi, 2012; Yang et al., 2013). We were therefore interested to see whether MdARF17 could bind to the same cis-element. We found the TGTCTC cis-element in the *MdHYL1* promoter (Supplemental Figure S11). Using a yeast one-hybrid (Y1H) assay, together with electrophoretic mobility shift assay (EMSA) and chromatin immunoprecipitation (ChIP)-quantitative polymerase chain reaction (qPCR) analyses, we confirmed that MdARF17 could bind to the promoter region of *MdHYL1* (Figure 6, E–G). In addition, we found that under control or drought conditions, *MdHYL1* was negatively regulated by MdARF17, indicating a potential role for MdHYL1 in apple drought tolerance (Figure 6H).

MdHYL1 is a positive regulator of drought tolerance and adventitious root development

MdHYL1 was induced by drought in a manner similar to that of Mdm-miR160 (Supplemental Figure S12A), further suggesting that MdHYL1 has a role in drought response. We generated *MdHYL1* transgenic plants with either enhanced or reduced *MdHYL1* expression levels (Supplemental Figure S12, B and C). When exposed to short-term drought stress, *MdHYL1* OE plants were more tolerant to drought and had a higher survival rate compared with GL-3 plants (Figure 7, A and B). Under drought stress, *MdHYL1* OE plants also had a lower ion leakage rate (Figure 7C). However, in contrary with *MdHYL1* OE plants, *MdHYL1* RNAi plants were more sensitive to short-term drought stress, with a lower survival

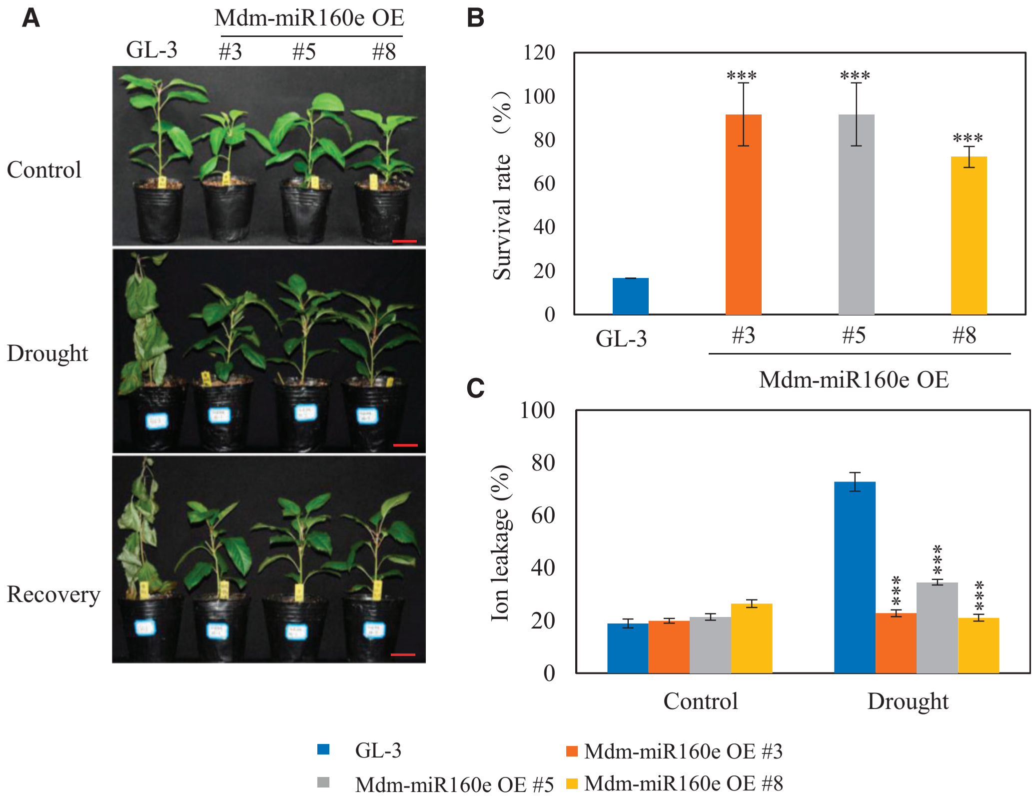

Figure 5 Drought stress tolerance of Mdm-miR160e OE plants. A, Drought tolerance of Mdm-miR160e OE plants. Two-month-old plants were transferred to soil for additional 1 month, and water was then withheld for 16 d until VWC decreased to 0. During the treatment, GL-3 plants were supplemented with water to maintain the same VWC with transgenic plants. Bars = 3 cm. B, Survival rate of plants in (A). C, Electrolyte leakage of Mdm-miR160e OE plants. Plants were exposed to drought until VWC decreased to 5%, and leaves were collected for an ion leakage assay. Error bars indicate SD (*n* = 15 in (E)), 15 plants were used and every 5 plants was a biological replication. One-way ANOVA (Tukey's test) was performed and statistically significant differences were indicated by ***$P < 0.001$.

rate and a higher ion leakage rate (Figure 7, D–F). These results suggest that MdHYL1 is a positive regulator of apple drought tolerance.

We also examined adventitious root development of *MdHYL1* transgenic plants. *MdHYL1* RNAi plants had reduced adventitious root length and lower adventitious root numbers, whereas *MdHYL1* OE plants had greater adventitious root length and higher adventitious root numbers (Supplemental Figure S13, A–C), suggesting that MdHYL1 is a positive regulator of adventitious root development. Moreover, after prolonged drought stress, *MdHYL1* OE plants had higher dry weight of roots and root-to-shoot ratio while *MdHYL1* RNAi plant had lower root dry weight and root-to-shoot ratio than GL-3 plants (Supplemental Figure S14, A–D).

MdHYL1 regulates biogenesis of drought-responsive miRNAs and mRNAs, including Mdm-miR160

Arabidopsis HYL1 is required for miRNA biogenesis and processing (Manavella et al., 2012; Cho et al., 2014). *hyl1* mutant plants show developmental defects, such as curly leaves, short stature, and narrower cauline leaves (Lu and Fedoroff, 2000; Vazquez et al., 2004). We also found that both *MdHYL1* OE and RNAi plants were dwarfed (Supplemental Figure S15). *MdHYL1* RNAi plants had slightly curly leaves, smaller leaf area, shorter leaf width, and shorter leaf length (Supplemental Figure S16). However, the leaf length/width ratio of *MdHYL1* RNAi and GL-3 plants was similar (Supplemental Figure S16). Leaf area, leaf width, leaf length, and leaf length/width ratio of *MdHYL1* OE plants did not differ significantly from those of GL-3 plants (Supplemental Figure S17).

We next examined the expression of miRNAs that function in drought tolerance (Yang et al., 2013; Chen et al., 2015a, 2015b; Hajyzadeh et al., 2015; Li et al., 2017). Under control conditions, MdHYL1 positively regulated the expression of all Mdm-miRNAs we detected. Under drought conditions, expression of Mdm-miR156a p5, Mdm-miR160, Mdm-miR164c p5, and Mdm-miR396 p5 was reduced in *MdHYL1* RNAi plants, whereas expression of Mdm-miR393g p5 and Mdm-miR398 was increased (Supplemental Figure S18).

To understand whether MdHYL1 can regulate the expression of mRNAs in response to drought, we performed a transcriptome analysis of *MdHYL1* RNAi plants under drought stress. The results showed that 238 differentially expressed genes were upregulated in the *MdHYL1* RNAi plants under drought, whereas 168 genes were downregulated (Supplemental Datasets S5 and S6). Among the

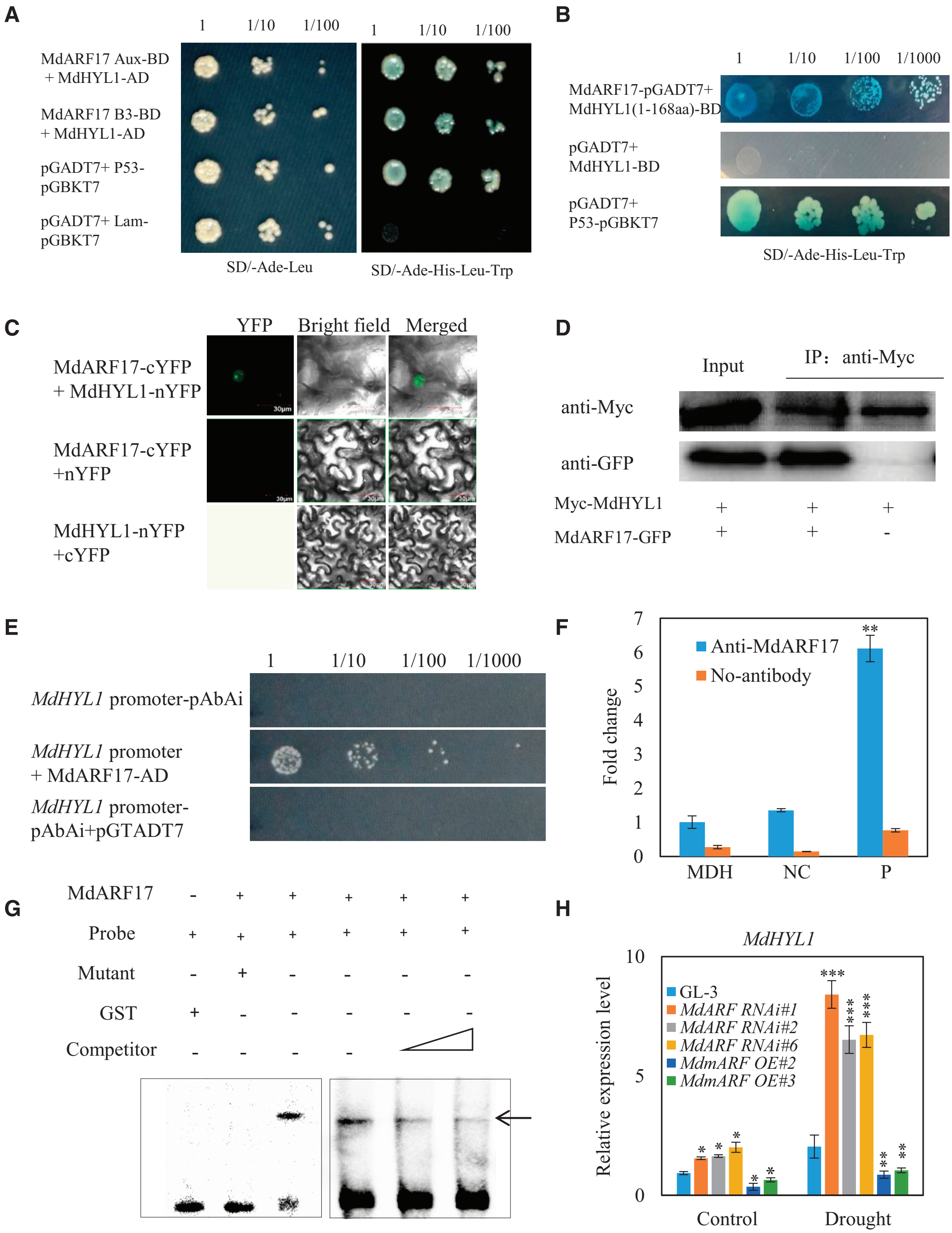

Figure 6 MdHYL1 interacts with MdARF17 and is a target of MdARF17. A and B, Y2H assay of MdHYL1 and MdARF17. Truncated MdARF17-B3 (107–240 aa), MdARF17-Aux (252–367 aa), and MdHYL1 (1–168 aa) which lack self-activation activities were cloned into pGBKT7. C, BiFC analysis of MdHYL1 and MdARF17. An argon laser excitation of 488 nm (intensity 1%) was examined at the wavelength of 492–537 nm to detect YFP signal. D, Co-IP analysis of MdHYL1 and MdARF17. E, Y1H assay of the *MdHYL1* promoter and MdARF17. F, ChIP-qPCR showing binding of MdARF17 to the *MdHYL1* promoter. MDH serves as the reference gene, NC is the negative control and Promoter contains the TGTCTC cis-element in the *MdHYL1* promoter (−534 bp to −437 bp). G, EMSA assay showing the interaction of MdARF17 protein and *MdHYL1* promoter. H, *MdHYL1* expression in *MdARF17* RNAi or *MdmARF17* OE plants under control and drought conditions. Error bars indicate SD (n = 3). One-way ANOVA (Tukey's test) was performed and statistically significant differences were indicated by $^{*}P < 0.05$; $^{**}P < 0.01$; $^{***}P < 0.001$.

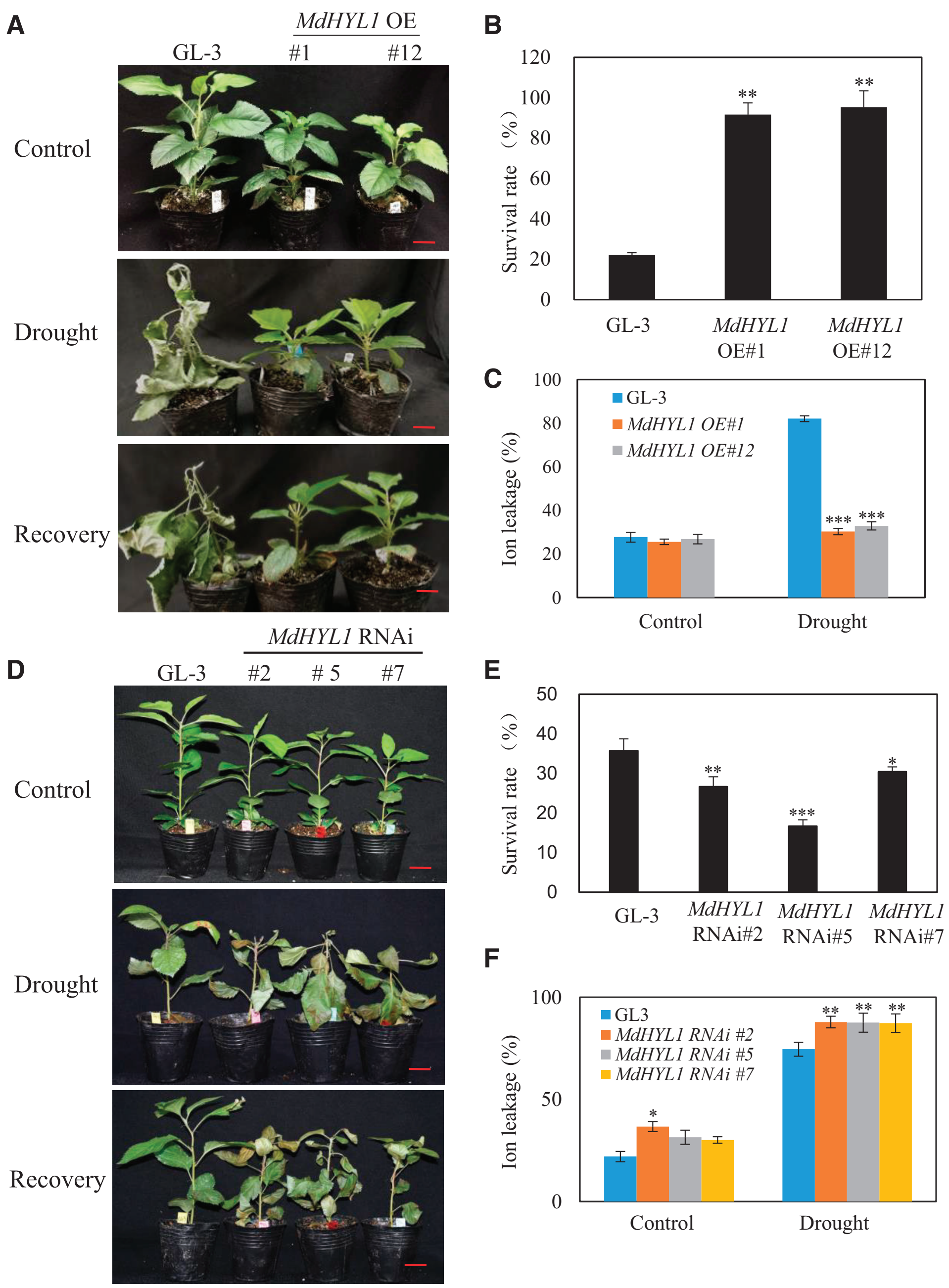

Figure 7 *MdHYL1* plays a positive role in apple drought tolerance. A, Drought tolerance of *MdHYL1* OE plants. Two-month-old plants were transferred to soil for additional 1 month, and water was then withheld for 16 d until VWC decreased to 0. During the treatment, GL-3 plants were supplemented with water to maintain the same VWC with transgenic plants. Bars = 3 cm. B, Survival rate of *MdHYL1* OE plants after drought. C, Electrolyte leakage of *MdHYL1* OE plants. Plants were exposed to drought until VWC decreased to 5%, and leaves were collected for an ion leakage assay. D, Drought tolerance of *MdHYL1* RNAi plants. Two-month-old plants were transferred to soil for additional 1 month, and water was then withheld for 14 d until VWC decreased to 0. During the treatment, GL-3 plants were supplemented with water to maintain the same VWC with transgenic plants. E, Survival rate of *MdHYL1* RNAi plants after drought. Bars = 3 cm. F, Electrolyte leakage of *MdHYL1* RNAi plants. Plants were exposed to drought until VWC decreased to 5%, and leaves were collected for an ion leakage assay. Error bars indicate SD (n = 15 in (B, E)), 15 plants were used and each 5 plants was a biological replication). One-way ANOVA (Tukey's test) was performed and statistically significant differences were indicated by $^{**}P < 0.01$; $^{***}P < 0.001$.

differentially expressed genes, the expression of drought-positive regulators, such as Autophagy-related protein 8i (*MdATG8i*), Abscisic acid receptor PYL4-like (*MdPYL4-like*), and *MdWRKY71*, was reduced in the *MdHYL1* RNAi plants under drought stress. However, the expression of a drought-negative regulator, WRKY DNA-binding protein 70 (*MdWRKY70*), was increased in the *MdHYL1* RNAi plants under drought (Supplemental Figure S19). Interestingly, we found 29 genes that were regulated by both MdARF17 and MdHYL1 (Supplemental Tables S1 and S2). Among the 29

DEGs, the drought-positive factors *MdWRKY71* and *MdEXP4* were upregulated in the *MdARF17* RNAi plants under drought but downregulated in the *MdHYL1* RNAi plants. Two genes, L-ascorbate oxidase (*MdLAO*) and WAT1-related protein (*MdWAT1*), were downregulated in the *MdARF17* RNAi plants but upregulated in the *MdHYL1* RNAi plants (Figure 3E; Supplemental Figures S19 and S20). Less overlapping DEGs in *MdHYL1* RNAi and *MdARF17* RNAi plants under drought stress indicate that MdHYL1 and MdARF17 might not be in the same pathway in regulating mRNA expression in response to drought stress.

MdARF17 negatively regulates the abundance of miRNAs under control and drought conditions

We also noticed that the *MdARF17* RNAi plants were shorter and had smaller leaf area, leaf length, and width compared with GL-3 plant leaves (Supplemental Figure S21), whereas *MdmARF17* OE plants did not differ from nontransgenic plants in leaf phenotype (Supplemental Figure S22).

Because MdARF17 interacts with MdHYL1 and negatively regulates *MdHYL1* expression under control and drought conditions (Figure 6), and because we observed similar developmental phenotypes in the *MdHYL1* RNAi and *MdARF17* RNAi plants, we next investigated whether MdARF17 could mediate the biogenesis of miRNAs under control and drought conditions. Using stem–loop RT-qPCR analysis, we found that MdARF17 negatively regulated the expression of miRNAs, including Mdm-miR399b p3, Mdm-miR156a p5, Mdm-miRn249, Mdm-miR398, Mdm-miR396 p5, Mdm-miR166c p5, Mdm-miR160, Mdm-miR164c p5, and Mdm-miR399b p5 under control conditions (Figure 8A), suggesting a role for MdARF17 in the negative regulation of miRNA biogenesis. We also found that MdARF17 could negatively regulate the expression of some drought-responsive miRNAs, including Mdm-miR156a p5, Mdm-miR160, Mdm-miR164c p5, Mdm-miR166c p5, Mdm-miR396 p5, Mdm-miR408a, Mdm-miR399 p5, and Mdm-miR398 (Figure 8B; Supplemental Figure S23).

MdHYL1 acts downstream of MdARF17 in regulating Mdm-miRNAs abundance and drought tolerance

To investigate the genetic relationship between MdHYL1 and MdARF17 in miRNA biogenesis, we transiently knocked down *MdHYL1* in the *MdARF17* RNAi plants. We found that Mdm-miR160 abundance was reduced by 75% in the *MdARF17* RNAi plants after *MdHYL1* was knocked down (Supplemental Figure S24). In addition, transcripts of Mdm-miR156a p5, Mdm-miR166c p5, Mdm-miR164c p5, Mdm-miR396 p5, and Mdm-miR398 were also reduced in the double RNAi plants (Supplemental Figure S24), indicating that MdARF17 regulates these Mdm-miRNAs through MdHYL1.

To further investigate the genetic relationship of MdHYL1 and MdARF17 in apple Mdm-miRNA biogenesis and drought tolerance, we obtained stably transformed apple calli with reduced expression of *MdARF17* in *MdHYL1* apple calli (Supplemental Figure S25). Under control conditions, knocking down *MdARF17* in *MdHYL1* RNAi calli did not alter or slightly rescued transcripts of Mdm-miR156a p5, Mdm-miR160, Mdm-miR164c p5, and Mdm-miR398. After polyethylene glycol (PEG) treatment, abundance of Mdm-miR156a and Mdm-miR164c p5 in *MdARF17* RNAi/*MdHYL1* RNAi was comparable to that of *MdHYL1* RNAi calli, whereas reduced transcripts of Mdm-miR160 in *MdHYL1* RNAi calli were slightly recovered when *MdARF17* expression was repressed. In addition, suppressed *MdARF17* decreased Mdm-miR398 abundance in *MdHYL1* RNAi background after PEG treatment (Figure 9A). We also examined the tolerance of transgenic calli to PEG treatment. After PEG treatment for 15 d, the *MdHYL1* RNAi calli and *MdARF17* RNAi/*MdHYL1* RNAi calli displayed reduced tolerance to simulated drought stress, whereas *MdARF17* RNAi plants showed increased tolerance (Figure 9, B and C). Moreover, we produced transgenic apple calli with increased *MdARF17* in *MdHYL1* RNAi background (Supplemental Figure S25). We found that overexpressing *MdmARF17* did not alter the abundance of Mdm-miR156a p5, Mdm-miR160, Mdm-miR164c p5, and Mdm-miR398 in *MdHYL1* RNAi calli, either under control or after PEG treatment (Figure 9D). In addition, both *MdmARF17* OE and *MdHYL1* RNAi calli were more sensitive to simulated drought stress, as compared with the wild-type (WT). However, the tolerance of *MdmARF17* OE/*MdHYL1* RNAi apple calli was comparable to that of *MdHYL1* RNAi calli, further indicating that MdHYL1 acts downstream of MdARF17 in apple drought tolerance and Mdm-miRNAs biogenesis (Figure 9, E and F).

Finally, we transformed *MdHYL1* RNAi into the roots of *MdARF17* RNAi plants with a hairy root transformation system mediated by *Agrobacterium rhizogenes*, and induced hairy roots of *MdHYL1* RNAi/*MdARF17* RNAi. GL-3 roots transformed with the empty vector served as a control. After removal of nontransgenic roots, plants were grown in pots in a growth chamber. When treated with short-term drought stress, *MdHYL1* RNAi/*MdARF17* RNAi plants had a lower survival rate compared with the control plants (Supplemental Figure S26), further suggesting that MdHYL1 is epistatic to MdARF17 in apple drought tolerance.

Overall, the above results suggest that MdARF17 regulates steady state levels of Mdm-miRNAs and drought tolerance by regulating *MdHYL1*, at least in part.

Mdm-miR160 can move from scion to rootstock and promotes drought tolerance in the rootstock

Some miRNAs are mobile (Melnyk et al., 2011). Grafting has been widely used to facilitate flowering, dwarfing, and stress tolerance in apple trees (Wang et al., 2019). In situ hybridization analysis revealed that Mdm-miR160 was abundant in vascular tissues, including xylem and phloem (Figure 10A). To understand whether Mdm-miR160 moves between scion and rootstock, we performed micro-grafting analysis using GL-3 and *MdHYL1* RNAi plants as the rootstock and the scion mutually. We found that when GL-3 plants were used

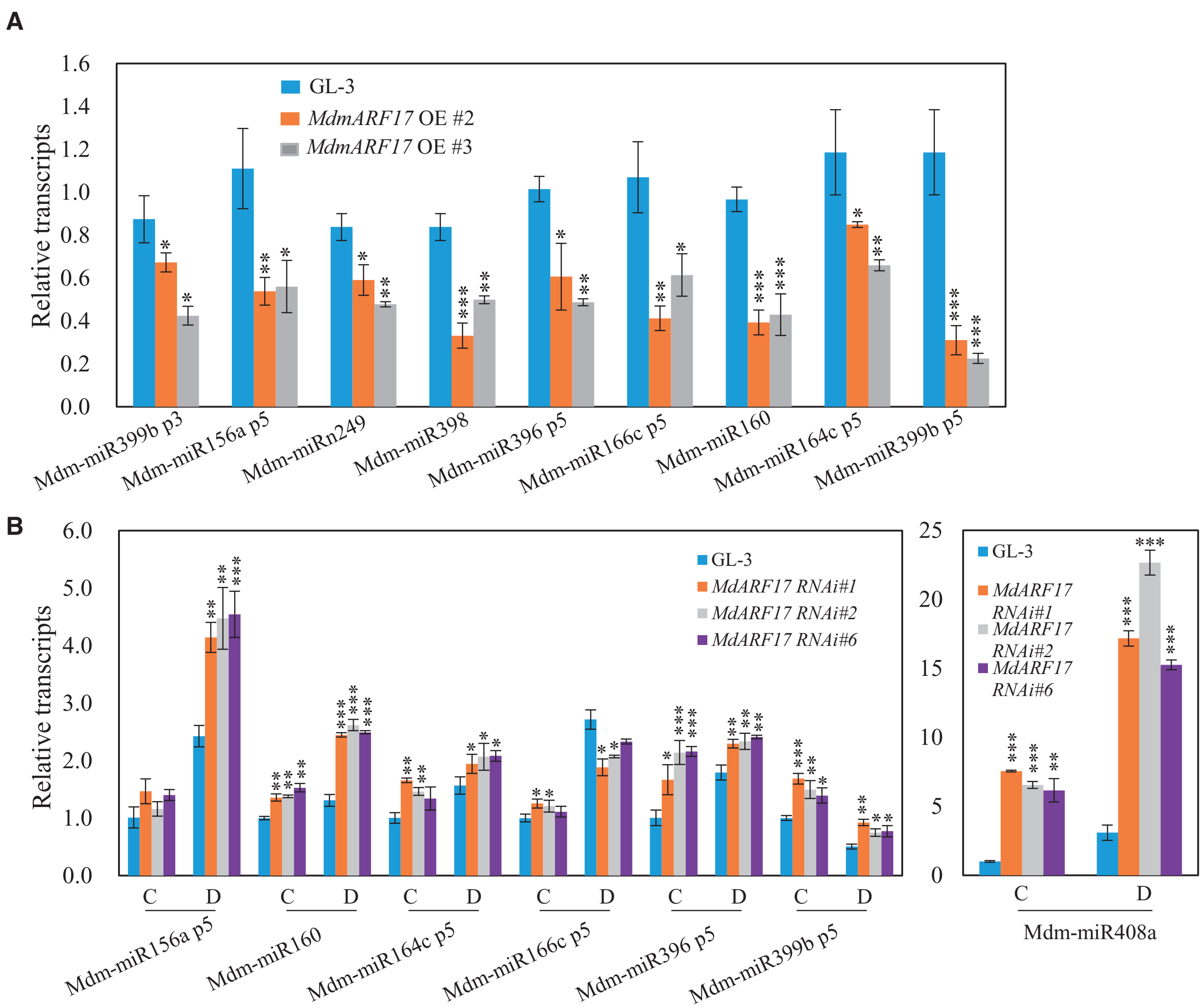

Figure 8 MdARF17 regulates transcripts of Mdm-miRNAs. A, Transcripts of Mdm-miRNAs in *MdmARF17* OE plants. B, Transcripts of Mdm-miRNAs in *MdARF17* RNAi plants under control and drought conditions. Data are the means ± SD (n = 3). C, control. D, drought. One-month-old plants were exposed to drought until VWC decreased to 5%, and leaves were collected for total RNA extraction and stem–loop RT-qPCR analysis. One-way ANOVA (Tukey's test) was performed and statistically significant differences were indicated by *$P < 0.05$; **$P < 0.01$; ***$P < 0.001$.

as the scion, Mdm-miR160 transcripts were higher in the scion of GL-3/GL-3 than in that of GL-3/*MdHYL1* RNAi (GL-3 plants were grafted onto GL-3 plants). When *MdHYL1* RNAi plants were used as the rootstock, Mdm-miR160 transcript abundance was higher in the rootstock of GL-3/*MdHYL1* RNAi (GL-3 plants were grafted onto *MdHYL1* RNAi plants) than in that of *MdHYL1* RNAi/*MdHYL1* RNAi (Supplemental Figure S27), indicating that Mdm-miR160 might move from the scion to the rootstock.

To further verify the mobility of Mdm-miR160 from the scion to the rootstock, we grafted tobacco (*Nicotiana tabacum*) plants onto tomato (*S. lycopersicum*) plants (Figure 10B). The tobacco genome contains one Nta-miR160d which is not present in tomato genome. After grafting for 7 d, we were able to detect the Nta-miR160d sequence in tomato plants (Figure 10C; Supplemental Figure S28). Consistently, transcripts of Nta-miR160d were more abundant in the rootstock of tobacco/tomato (tobacco plants were grafted onto tomato plants) than in the rootstock of self-grafted tomato plants (Figure 10D).

To further support the notion that miR160 could move from the scion to the rootstock in apple, we grafted Mdm-miR160e OE plants (as the scion) onto WT GL-3 plants (as the rootstock) (Mdm-miR160e OE/GL-3), using self-grafted GL-3/GL-3 as controls. In situ hybridization analysis displayed that more Mdm-miR160 could be detected in the graft union after 4 d of grafting and in the rootstock after 7 d of grafting (Figure 10E). We also examined the expression of Mdm-miR160 and MdARF17 in Mdm-miR160e OE/GL-3 and GL-3/GL-3 plants. We found that after 7 d of grafting, Mdm-miR160 transcripts in GL-3 rootstock of Mdm-miR160e OE/GL-3 were increased, while MdARF17 expres-

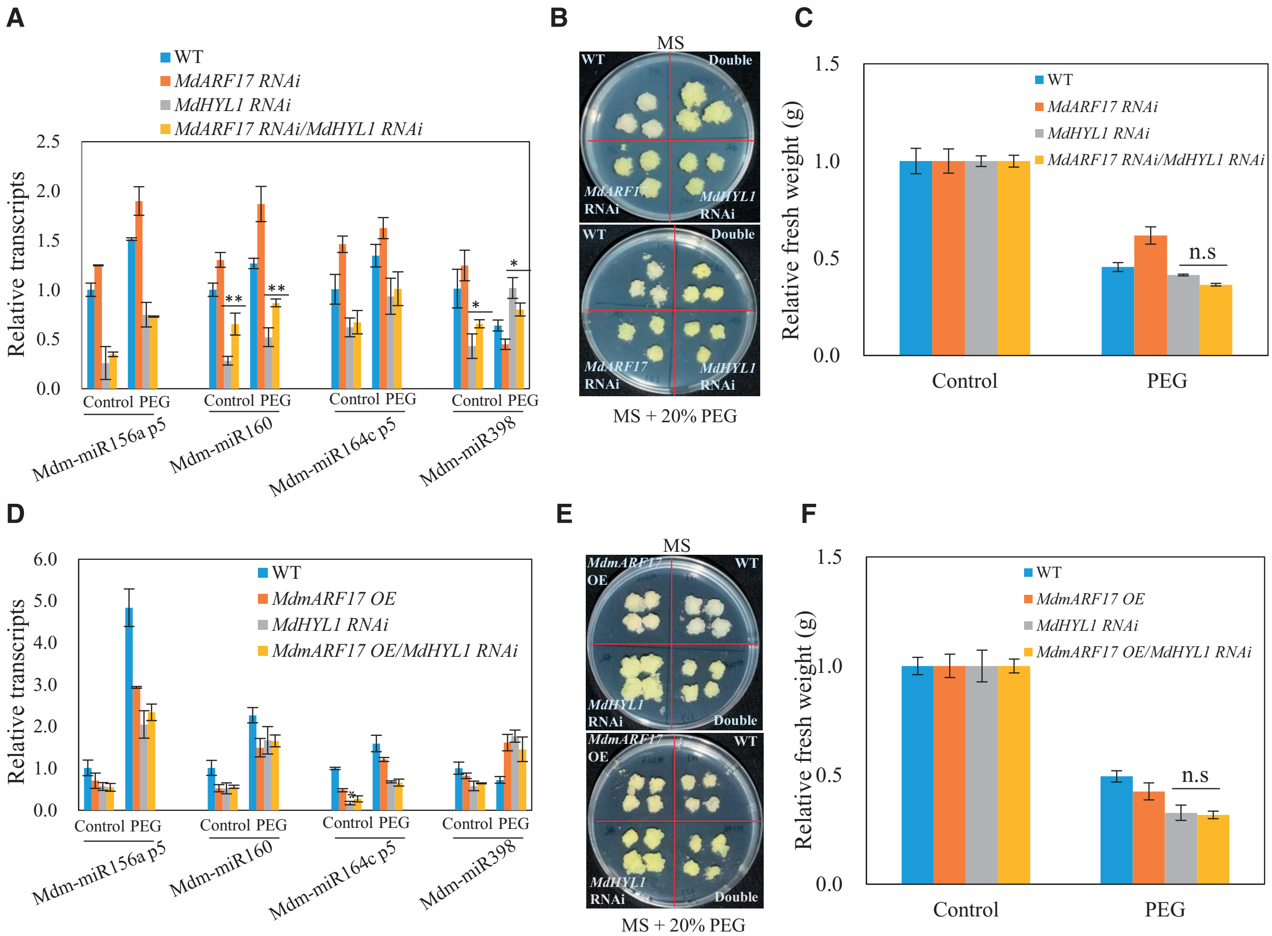

Figure 9 MdHYL1 acts downstream of MdARF17 in simulated drought tolerance and miRNA biogenesis. A–C, Mdm-miRNAs abundance and simulated drought tolerance of WT, *MdARF17* RNAi, *MdHYL1* RNAi, and *MdMARF17* RNAi/*MdHYL1* RNAi calli under control and PEG treatment. Double in (B), repressing *MdARF17* in *MdHYL1* RNAi calli. D–F, Mdm-miRNAs abundance and simulated drought tolerance of WT, *MdmARF17* OE, *MdHYL1* RNAi, and *MdmMARF17* OE/*MdHYL1* RNAi calli under control and PEG treatment. Double in (E), overexpressing *MdmARF17* in *MdHYL1* RNAi calli. Calli were grown son MS medium or MS medium supplemented with 20% (w/v) PEG (−0.7 MPa) for 15 d. Data are means ± SD (n = 3). One-way ANOVA (Tukey's test) was performed and statistically significant differences were indicated by *P < 0.05; **P < 0.01. ns, not significant.

sion was decreased (Figure 10F). Taken together, these results suggest that miR160 can move from the scion to the rootstock in apple and tobacco plants.

Considering that miR160 could move from the scion to the rootstock, we examined the biological roles of Mdm-miR160 in root development of the rootstock using the grafted plants of Mdm-miR160e OE/GL-3 and GL-3/GL-3. We found that Mdm-miR160e OE/GL-3 had more adventitious roots and greater root length than that of GL-3/GL-3 plants (Figure 11, A–C). Since root system plays critical roles in plant drought tolerance (Kim et al., 2020), we then evaluated drought tolerance of the grafted plants. When exposed to drought, Mdm-miR160e OE/GL-3 plants were more tolerant than GL-3/GL-3 plants (Figure 11, D and E). Moreover, Mdm-miR160e OE/GL-3 plants had greater root length after drought than GL-3/GL-3 plants (Figure 11F). These data suggest that mobile Mdm-miR160 from the scion improves root development and drought tolerance of the rootstock.

Discussion

Drought stress is a complex trait regulated by multiple factors. Identifying key components of this regulatory network is crucial to genetically modify the drought tolerance of perennial trees. This is particularly true for apple, as it is often self-incompatible and has a long juvenile phase, typically extending the breeding process to 10 years or more. In this study, we confirmed that MdARF17 negatively influences apple drought tolerance: the *MdARF17* RNAi plants were more tolerant, whereas the *MdmARF17* OE plants were more sensitive (Figure 2; Supplemental Figures S4 and S5). Conversely, the regulatory miRNA of *MdARF17*, Mdm-miR160, is a positive regulator of apple drought stress tolerance (Figure 5; Supplemental Figure S8). The direct target of MdARF17, *MdHYL1*, also has a positive role in apple drought tolerance (Figure 7; Supplemental Figures S14 and S15). Hence, the Mdm-miR160–MdARF17–MdHYL1 module is crucial for apple drought tolerance and may be used in the breeding of drought-tolerant cultivars.

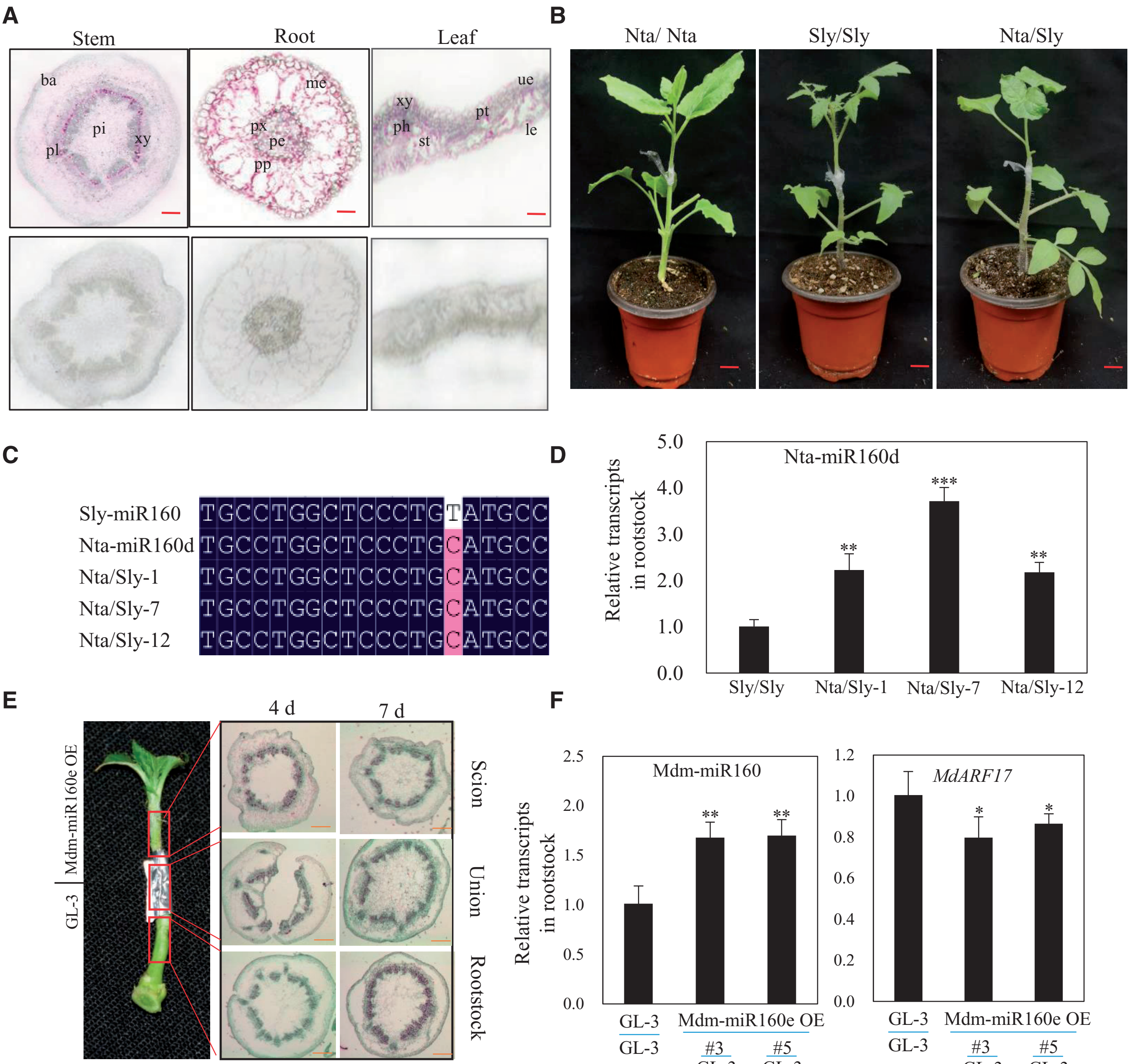

Figure 10 miR160 can move from the scion to the rootstock. A, Transcripts of Mdm-miR160 in roots, stems, and leaves using an in situ hybridization assay. Ba, bark. Xy, xylem. Ph, phloem. Pi, pith. Px, primary xylem. Pp, primary phloem. Me, mediopellis. Pe, pericycle. Bars = 80 µm. B, Nta/Nta, tobacco plants were grafted onto tobacco plants. Sly/Sly, tomato plants were grafted onto tomato plants. Nta/Sly, tobaccos were grafted onto tomatoes. C, Detection of tobacco miR160 in tomato plants using the Nta/Sly plants. RNA was extracted and tobacco miR160d was cloned and sequenced. D, Transcripts of miR160 in tomato plants using the Sly/Sly and Nta/Sly plants. After grafting for 7 d, the phloem of the rootstock was used for RNA extraction, followed by stem-loop RT-qPCR. E, Transcripts of Mdm-miR160 in micro-grafted plants. Left: a representative of micro-grafted plant. The scion (Mdm-miR160e OE plants) was grafted on the rootstock (GL-3 plants) for 4 and 7 d, and the stem fragments containing graft union, or 1 cm below or above the graft union were used to detect the transcripts of Mdm-miR160 by in situ hybridization. Bars = 80 µm. F, Transcripts of Mdm-miR160 or *MdARF17* in the stem of GL-3 rootstock when GL-3 or Mdm-miR160e OE plants were used as scion. Error bars indicate SD ($n = 3$). One-way ANOVA (Tukey's test) was performed, and statistically significant differences were indicated by *$P < 0.05$; **$P < 0.01$.

Auxin response factor (ARF) proteins in maize (*Zea mays* L.), longan (*Dimocarpus longan* L.), litchi (*Litchi chinensis* Sonn.), kiwifruit (*Actinidia chinensis*), and sugar beet (*Beta vulgaris* L.) contain three conserved regions: an amino-terminal DNA-binding domain (DBD) consisting of a B3-type DBD, middle region (MR), which is responsible for gene activation or repression, and carboxy-terminal dimerization domain (CTD; Wang et al., 2012; Zhang et al., 2019; Cui et al., 2020; Peng et al., 2020; Su et al., 2021). Previous studies revealed that full length or CTD domain of ARF proteins can interact with Auxin/Indole-3-Acetic Acid (Aux/IAA) proteins (Shen et al., 2010; Guilfoyle, 2015; Wu et al., 2018). For

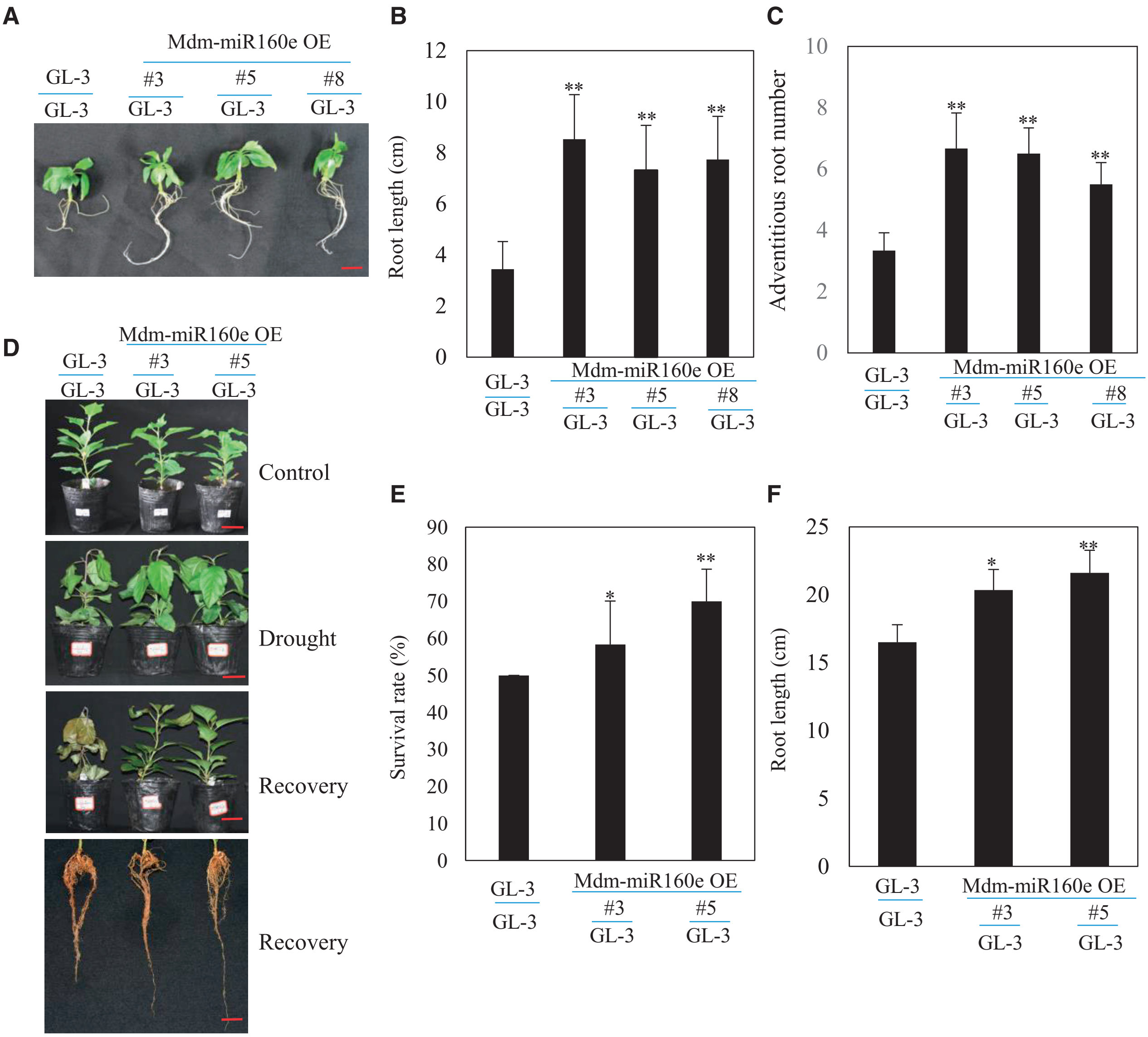

Figure 11 Mobile Mdm-miR160 from the scion promotes drought tolerance of the rootstock. A–C, Root morphology (A), root length (B), and adventitious root number (C) of the GL-3 rootstock when GL-3 or Mdm-miR160e OE plants were used as the scion. Bar = 3 cm. D, Drought tolerance of the grafted plants. Two-month-old plants were transferred to soil for additional 1 month, and water was then withheld for 16 d until VWC decreased to 0. During the treatment, GL-3/GL-3 plants were supplemented with water to maintain the same VWC with grafted plants. Bars = 3 cm. E and F, Survival rate (E) and root length (F) of the grafted plants after drought stress. Error bars indicate SD (n = 10 in (B, C, E and F)). One-way ANOVA (Tukey's test) was performed and statistically significant differences were indicated by $^{*}P < 0.05$; $^{**}P < 0.01$.

instance, Y2H analysis showed that the CTD or the MR + CTD of OsARF6, 11, 12, 16, 17, 19, and 25 interacts with 15 OsIAA proteins (Shen et al., 2010). Besides, ERF72 physically interacts with the CTD of ARF6 to regulate hypocotyl elongation in Arabidopsis (Liu et al., 2018a, 2018b). However, the DBD, but not the CTD, of OsARF17 interacts with RSV P2, a weak RNA silencing suppressor (Zhang et al., 2020). In our study, we found that the B3 domain (included in DBD) and the Aux domain (included in CTD) of MdARF17 can interact with MdHYL1 in yeast, the full length of MdARF17 can interact with MdHYL1 in tobacco by BiFC and CO-IP assays. Our results are consistent with previous studies (Zhang et al., 2020), further supporting that the B3 domain can also interact with other proteins.

As TFs, ARFs bind specifically to the Auxin Response Element TGTCTC in the promoters of multiple genes including Aux/IAAs, small auxin-up RNA family members, and Gretchen Hagen3 (Guilfoyle and Hagen, 2007; Luo et al., 2018), either enhancing or repressing their transcription (Hayashi, 2012; Luo et al., 2018). Arabidopsis ARF17 has previously been reported to bind the *MYB108* promoter to regulate anther dehiscence (Xu et al., 2019). In our study, we verified that MdARF17 directly and negatively regulates the expression of *MdHYL1* (Figure 6), as well as positively and negatively regulating the expression of additional genes under drought stress (Figure 3E). Under drought stress, MdARF17 negatively regulated drought-positive regulators including *MdWRKY71*, *MdTIP1-3like*, *MdLTP3*, *MdEXP4*, *MdACS*, *MdHSP70*, *MdMADS*, and *MdSFBB9*, as well as drought-positive miRNAs including Mdm-miR156a p5, Mdm-miR160, Mdm-miR164c p5, Mdm-miR166c p5, Mdm-miR396 p5, and Mdm-miR408a (Figures 3, E and 8). In addition, MdARF17 positively regulated expression of drought-repressing factors such as *MdLAO* and *MdWAT1* (Supplemental Figure S20). Drought tolerance regulation by MdARF17 can therefore be attributed to the negative and positive regulation of drought-responsive mRNAs and miRNAs.

In our study, MdARF17 directly binds to the promoter of *MdHYL1*. Genetic analysis suggested that MdARF17 regulates miRNA biogenesis and drought tolerance via *MdHYL1*. Though our RNA-seq data revealed that MdARF17 activates expression of drought-responsive genes, the RT-qPCR results showed that MdARF17 negatively regulates *MdHYL1* expression in response to drought. In plants, pri-miRNAs are processed by the dicing complex with DCL1, HYL1, and SE as the core components, to yield mature miRNA/miRNA* duplexes. Therefore, it is possible that MdARF17 may influence the specificity of MdDCL1/MdHYL1/MdSE complex to specific pri-miRNAs by regulating *MdHYL1* expression.

The drought tolerance conferred by MdHYL1 appeared to arise from the positive regulation of drought-responsive miRNAs, including Mdm-miR156a p5, Mdm-miR160, Mdm-miR164c p5, and Mdm-miR396 p5, as well as the negative regulation of Mdm-miR393g p5 and mdm-miR398 (Supplemental Figure S18). In addition, MdHYL1 also regulated the transcript levels of drought-responsive mRNAs. For example, *MdATG8i*, *MdPYL4-like*, *MdEXP4*, and *MdWRKY71* were downregulated in the *MdHYL1* RNAi plants under drought, whereas *MdWRKY70* and *MdLAO* were upregulated (Supplemental Figures S19 and S20). *MdATG8i* and homologs of *MdPYL4-like*, *MdEXP4*, and *MdWRKY71* have previously been identified as drought-positive factors, and homolog of *MdWRKY70* as a negative factor (Pizzio et al., 2013; Huang et al., 2019). Therefore, regulation of both miRNAs and mRNAs contribute to the positive effect of MdHYL1 on drought stress response.

miRNAs are important players in plant drought response. Although miRNAs are often conserved across plant species, they may have opposite roles under drought conditions in different species. OE of Osa-miR393a in creeping bentgrass (*A. stolonifera* L.) conferred improved drought stress tolerance associated with reduced stomatal density and denser cuticles (Zhao et al., 2019). However, OE of Osa-miR393 in rice reduced drought stress tolerance (Xia et al., 2012), and transgenic Arabidopsis plants that overexpressed Osa-miR393 exhibited greater sensitivity to osmotic stress (Chen et al., 2015a, 2015b). Osa-miR408 and ath-miR408 OE plants exhibited decreased drought tolerance (Ma et al., 2015; Sun et al., 2018), whereas higher expression of miR408 in chickpea led to increased drought tolerance (Hajyzadeh et al., 2015). Consistent with these findings, miRNAs from different plant species showed different patterns in response to drought stress (Ferdous et al., 2015). Under drought stress, miR160 is downregulated in rice and tomato but upregulated in other crops such as wheat, peach, and cowpea (Ferdous et al., 2015), indicating that miR160 can play different roles in response to drought. Our results clearly demonstrated that transgenic apple plants carrying drought-inducible Mdm-miR160 were more tolerant to drought stress (Figure 5). Considering the diverse expression patterns of miR160 in different plant species, we speculate that its functions under drought may differ among species.

Some TFs are targeted by their cognate miRNAs, thereby regulating MIR expression in a complex model of regulatory loops. In Arabidopsis, the miR160 family contains three members (ath-miR160a, b, and c), and ARF17 is one of its targets. To gain insight into the effects of ARF17 on the regulation of its own regulatory miRNA, Gutierrez et al. (2009) analyzed the expression of pri-miRNAs (pri-miR160a, pri-miR160b, and pri-miR160c) and the steady-state levels of mature miR160. In the cited study, pri-miR160 levels in *ARF17* OE plants did not differ from those in WT plants; however, mature miR160 levels were lower in *ARF17* OE plants, suggesting that ARF17 regulates the abundance of miR160 but not pri-miR160 (Gutierrez et al., 2009). In our study, mature mdm-miR160 levels were increased in the *MdARF17* RNAi plants but decreased in the *MdmARF17* OE plants (Figure 8), consistent with previous research. In addition, *MdHYL1* was a direct target of MdARF17 (Figure 6), and MdHYL1 positively regulated the abundance of Mdm-miR160 (Supplemental Figure S18). Furthermore, when *MdHYL1* was knocked down in the *MdARF17* RNAi plants, the abundance of Mdm-miR160, Mdm-miR156a p5, Mdm-miR164c p5, Mdm-miR166c p5, Mdm-miR396 p5, and Mdm-miR398 was decreased (Supplemental Figure S24), suggesting that MdARF17 regulates miRNA levels through MdHYL1. Therefore, the Mdm-miR160–MdARF17–MdHYL1 module may form a positive feedback regulatory loop.

Arabidopsis ARF17 does not regulate the levels of ARGONAUTE 1, *SE*, *HYL1*, and HUA ENHANCER 1 (Gutierrez et al., 2009), suggesting that ARF17 does not affect the main miRNA biogenesis pathway. However, we identified MdHYL1 as an interacting protein and a direct target of MdARF17 (Figure 6) and found that MdHYL1 positively regulated steady state levels of mature miRNAs (Supplemental Figure S18). In addition, MdARF17 negatively regulated the transcript amounts of *MdHYL1* and miRNAs (Figures 6, H and 8). Indeed, both MdARF17 and MdHYL1 regulated a number of the same miRNAs, including Mdm-miR156a p5, Mdm-miR160, Mdm-miR164c p5, Mdm-miR166c p5, Mdm-miR396 p5, and Mdm-miR398 (Figure 8; Supplemental Figures S18 and S23). Furthermore, knocking down *MdARF17* in the *MdHYL1* RNAi plants slightly rescued the abundance of mature miRNAs, while overexpressing MdARF17 did not alter miRNAs abundance, including

Mdm-miR160 (Figure 9; Supplemental Figure S25), further supporting the possibility that *MdHYL1* is a downstream target of MdARF17. Therefore, ARF17 and MdARF17 may have different mechanisms for regulating miRNA levels in Arabidopsis and apple.miR160-mediated ARF17 regulates multiple aspects of plant development. OE of the miR160-resistant version of ARF17 results in ARF17 overaccumulation and dramatic developmental defects in leaves and embryos, as well as sterility. In particular, these transgenic plants display reduced primary root length and lateral root number (Mallory et al., 2005). In another research, *ARF17* OE plants show reduced adventitious root number, whereas miR160a and miR160c OE plants display an increased number (Gutierrez et al., 2009). Interestingly, poplar overexpressing miR160a shows reduced root length, and Arabidopsis expressing poplar *35S: PeARF17.1* and *35S:mPeARF17.2* shows an increased number of adventitious roots. In addition, adventitious roots are shorter in *mPeARF17.2* transgenic Arabidopsis (Liu et al., 2020). This study indicates that the miR160–ARF17 module in poplar is more complex. In our research, *MdARF17* RNAi plants and Mdm-miR160e OE plants had greater root length and more adventitious roots (Supplemental Figures S4 and S8). In contrast, *MdmARF17* transgenic plants displayed fewer adventitious roots and reduced root length (Supplemental Figure S4). Our results therefore suggested that the regulation of adventitious root development may be conserved between Arabidopsis and apple.

Root system architecture plays a critical role in plant drought tolerance, as roots are essential for absorption of water and nutrients from the surrounding environment. The root length and adventitious root number are considered to be the most promising traits for plant drought tolerance (Li et al., 2015). In addition, plants maintain the growth and development under drought stress via adjusting their secondary metabolites such as osmolytes and antioxidants to protect cell membranes from damage (Batool et al., 2020; Ozturk et al., 2021). In our study, drought-tolerant transgenic plants, including Mdm-miR160e OE, *MdARF17* RNAi, and *MdHYL1* OE plants, had longer adventitious roots and more root biomass under drought stress, as compared with GL-3 plants (Supplemental Figures S4, S5, S8, S13, and S14). The root length of these plants increased by 27%, 21%, and 47% as compared with GL-3 in response to drought stress, respectively (Supplemental Figures S4, S8, and S13). The root biomass of these plants increased by 9.9%, 9.8%, and 8.4%, respectively (Supplemental Figures S5, S8, and S14). However, 21%–47% increase in root length and 8.4%–9.9% increase in root biomass of these plants may not be sufficient to explain their survival in drought conditions. Therefore, other factors, such as osmolytes and antioxidants may contribute to the drought tolerance of these plants.

In this study, mdm-miR160e OE, *MdARF17* RNAi, and *MdHYL1* OE plants had increased adventitious root length, whereas *MdmARF17* OE and *MdHYL1* RNAi plants had reduced root length. After prolonged drought stress, these transgenic plants had greater root dry weight and root-to-shoot ratio than the nontransgenic plants (Supplemental Figures S5, S8 and S14).

In conclusion, our results provide evidence of a positive feedback regulatory loop, Mdm-miR160-MdARF17-MdHYL1, that influences apple drought tolerance and has a role in adventitious root development (Supplemental Figure S29).

Materials and methods

Plant materials, growth conditions, and stress treatment

For gene cloning and expression in response to drought stress, "Golden Delicious" (*M.* × *domestica*) apple trees grown in a greenhouse were used for RNA extraction. One-month-old tissue-cultured GL-3 (selected from progenies of *M.* × *domestica* "Royal Gala"; Dai et al., 2013) plants were used for *Agrobacterium*-mediated genetic transformation as described previously (Xie et al., 2018).

GL-3 and transgenic plants were rooted in 1/2 murashige & skoog (MS) medium (2.215 g L^{-1} MS salts, 20 g L^{-1} sucrose, and 7 g L^{-1} agar, pH 5.8) supplemented with 0.5 mg L^{-1} indole-3-butytric acid and 0.5 mg L^{-1} IAA under dark conditions for 3 d and then held under long-day conditions (14-h light/10-h dark) for additional 2 months. Plants were then transplanted into soil and grown in a light growth chamber with 60% humidity under long-day conditions.

After plants were transplanted into soil and grown for 1 month, the short-term drought treatment was performed. Plants were then withheld with water until volumetric water content (VWC) decreased to 0. After additional 2 d, the status of intact plants was photographed, and plants were recovered under normal conditions. Survival rate was counted after re-watering for 5 d. During the treatment, the GL-3 plants were supplemented with water to maintain the same VWC with transgenic plants. VWC was detected with the soil moisture meter TDR350 (Spectrum America Inc., Boca Raton, FL, USA). An initial VWC of 50% was required before the treatment. Fifteen plants were used for each treatment of each line.

For the long-term drought treatment, 3-month-old GL-3 and transgenic plants were transferred to pots (30 × 18 cm) for an additional month in a greenhouse. Then plants of each line were divided into a well-watered group and long-term drought group, with 12 plants for each group of each line. Plants of the well-watered group were irrigated normally to maintain a field capacity of 75%–85%; plants of the long-term drought group were daily weighed and irrigated to maintain a field capacity of 45%–55%. The long-term drought stress treatment lasted 3 months. At the end of treatment, roots, leaves, and stems were collected for further analysis.

Measurement of ion leakage and leaf development parameters

Ion leakage experiments were carried out using leaves of GL-3 and transgenic plants before or after short-term drought stress when VWC decreased to 5%. Fully expanded leaves were punched into leaf discs with diameter of 50 mm. Two to three leaf discs were placed in a test tube containing 8 mL of deionized H_2O and stayed overnight. The ionic exudation rate was measured by a conductivity meter and followed by another measurement after the leaf discs were autoclaved. Electrolyte leakage was calculated as the percentage of the conductivity before autoclaving relative to that after autoclaving (Bhardwaj et al., 2014). Six leaves were used as one biological replicate, and six replicates were used for each treatment.

The leaf length, width, and length–width ratio of GL-3 and transgenic plants under control and drought conditions were measured by a leaf area scanner (Perfection V19, EPSON, Nagano, Japan).

Vector construction

For the RNAi construct, a 384-bp fragment of the *MdARF17* or a 368-bp segment of the *MdHYL1* coding sequence was amplified by PCR and cloned into the pK7GWIWG2D vector by a gateway technology (Invitrogen, Thermo Scientific, Waltham, MA, USA). *MdARF17* and *MdARF17*-like were all knocked down due to the high sequence similarity.

For the OE construct of Mdm-miR160e and *MdHYL1*, the pri-miR160e, or the coding region of *MdHYL1* was cloned into the vector pDONR222, and then to the destination vector pK2GW7 by a gateway technology (Invitrogen, Thermo Scientific, USA).

To generate OE vector of *MdmARF17*, five mutations were introduced by site-directed mutagenesis as described previously (Mallory et al., 2005).

Gene transformation

Agrobacterium tumefactions-mediated stable transformation with GL-3 as the background was carried out as described previously (Dai et al., 2013).

To knock down *MdHYL1* in *MdARF17* RNAi plants using the *A. rhizogenes* mediated hairy root transformation method, a 382-bp antisense fragment of *MdHYL1* was cloned into pGWB455 vector. Four-week-old tissue-cultured *MdARF17* RNAi plants were infected with the *A. rhizogenes* carrying antisense *MdHYL1*-pGWB455 for 30 min using a vacuum. Plants were then surface cleaned with sterile paper and cultured in rooting medium for additional 2 months. Fluorescence was observed with a microscope (Leica, Wetzlar, Germany), and roots without red fluorescent protein (RFP) fluorescence were removed. Plants were then transplanted into soil and grown in a growth chamber.

For the transient transformation in apple leaves, *A. tumefaciens* EHA105 carrying the *MdHYL1* RNAi construct was transiently transformed into the tissue-cultured leaves of 4-week-old *MdARF17* RNAi plants as described previously (Zhang et al., 2018a, 2018b). After 5 d of transformation, leaves were collected for gene expression detection.

Primers used are listed in Supplemental Table S3.

Y2H assay

For Y2H screen, full-length MdARF17, truncated MdARF17 containing B3 domain (107–240 aa) or Aux domain (252–367 aa) were cloned into pGBKT7 (for GAL4 BD fusion) to test self-activation activity. The truncated MdARF17 containing B3 domain which lacks the self-activation activity was further used as bait to screen the cDNA library.

To confirm the interaction between MdARF17 and MdHYL1 with Y2H, full-length MdHYL1 was cloned into pGADT7 (for GAL4 AD fusion), resulting in MdHYL1-pGADT7. MdHYL1-pGADT7 and truncated MdARF17 containing B3-pGBKT7 or Aux-pGBKT7 were co-transformed into Y2H Gold strain, followed by a selection on SD–Leu–Trp media and SD–Leu–Trp–His–Ade media. The positive clones were further tested with an x-gal assay.

Alternatively, truncated *MdHYL1* (1–168 aa) which contains the two RNA binding motif and lacks self-activation activity was cloned into pGBKT7, resulting in *MdHYL1* 168aa-pGBKT7. CDS of *MdARF17* was cloned into pGADT7, resulting in *MdARF17*-pGADT7. Both constructs (*MdHYL1* 168aa-pGBKT7 and *MdARF17*-pGADT7) were co-transformed into Y2H Gold strain, followed by a selection on SD–Leu–Trp media and SD–Leu–Trp–His–Ade media. The positive clones were further tested with an x-gal assay.

Primers used are listed in Supplemental Table S3.

BiFC assay

BiFC assay was performed as described (Walter et al., 2004; Yang et al., 2013). The coding sequence of *MdARF17* and *MdHYL1* were cloned into the pSPY-cEYFP and pSPY-nEYFP vector, respectively. The resulting constructs (*MdARF17*-cYFP and *MdHYL1*-nYFP) were transformed into *A. tumefaciens* strain C58C1, respectively, and then co-injected with *35S:P19* into 3-week-old tobacco leaves of *Nicotiana benthamiana*. Fluorescence signal was observed 3 d later with a confocal microscope (Nikon, Tokyo, Japan). An argon laser excitation of 488 nm (intensity 1%) was examined at the wavelength of 492–537 nm to detect yellow fluorescent protein (YFP) signal.

Primers used are listed in Supplemental Table S3.

Co-IP analysis

For Co-IP assays, the coding region of *MdHYL1* and *MdARF17* were cloned into the pEarlygate203 (with a Myc tag) and pEarlygate101 (with a green fluorescent protein [GFP] tag), respectively. The constructs (Myc-MdHYL1 and MdARF17-YFP) were transformed into *A. tumefaciens* C58C1, and then co-infiltrated into 3-week-old tobacco (*N. benthamiana*) leaves. Leaves were collected after 3 d and total protein was extracted in a protein extraction buffer containing 50-mM Tris–HCl, pH 7.5, 150-mM NaCl, 2-mM DTT, 10% (v/v) glycerol and 1% (v/v) NP-40, and protease inhibitor Cocktail (Thermo Scientific, USA). The anti-Myc antibody (1:500 dilution; Abmart, Shanghai, China) was co-incubated with crude protein extraction at 4°C overnight. The ChIP-Grade Protein A/G Magnetic Beads

(Thermo Scientific, USA) were added to the mixture and incubated for additional 4 h at 4°C. After wash for at least 3 times with protein washing buffer containing 50-mM Tris–HCl, pH 7.9, 150-mM NaCl, 2-mM EDTA, 5-mM DTT, 0.5% (v/v) NP-40, and 1 mM phenylmethylsulphonyl fluoride, the proteins in the immunoprecipitants were subjected to western blot detection with anti-Myc or anti-GFP antibody.

Primers used are listed in Supplemental Table S3.

Y1H assay

Y1H assay was performed according to the manufacturer's protocols of Matchmaker Gold Yeast One-Hybrid System Kit (Clontech, Mountain View, CA, USA). The coding region of *MdARF17* was cloned into pGADT7 to generate the *MdARF17*-pGADT7 vector. The *MdHYL1* promoter (−534 to −437 bp) was cloned into the pAbAi vector to generate pAbAi-bait plasmid which was then linearized and transformed into the Y1H Gold strain and selected with yeast standard deviation (SD) media lacking uracil but containing different concentrations of Aureobasidin A (AbA). The *MdARF17*-pGADT7 construct was transformed into the Y1H Gold strain containing pAbAi-bait and selected on SD-Ura media with AbA.

Primers used are listed in Supplemental Table S3.

ChIP-qPCR and EMSA analysis

ChIP-qPCR and EMSA were performed as described (Guan et al., 2013a, 2013b, 2014). To purify recombinant proteins from *Escherichia coli* cells, the coding region of Yang et al. (2013) *MdARF17* was cloned into the pMAL-C5X vector and transformed into *E. coli* strain BL21 (DE3). Expression and purification of MdARF17- maltose-binding protein fusion protein were performed using Amylose Resin (New England Biolabs, Ipswich, MA, USA) as described by Guan et al. (2014). To mutate the probe, the core binding site of TGTCTC was changed to AAAAAA.

The primers and probes are listed in Supplemental Table S3.

Micrografting assay

Micrografting was performed as described (Martin et al., 2009) with modifications. The tissue-cultured GL-3 and Mdm-miR160e OE plants cultured for 4 weeks were used as rootstocks or scions. Micrografting was carried out using peg grafting in a clean bench. The junctions were wrapped with the tinfoil and the grafted plants were cultured in the subculture medium. When the grafted junctions healed (∼10 d later), the phloem 1 cm below the graft junction was collected. Three biological repetitions were designed and each biological repetition has eight grafted plants. The combination of Mdm-miR160e OE scions onto GL-3 rootstocks was used to analyze the miRNA mobility and GL-3 scions onto GL-3 rootstocks were used as control. Tobacco plants were grafted onto tomato plants using a cut-in method. The phloem 1 cm below the union was collected for miR160 amplification and sequencing.

Subcellular localization and co-localization

Subcellular localization and co-localization were performed as described (Xie et al., 2018). The coding regions of *MdARF17* and *MdARF17*-like were amplified by PCR and cloned into the pEearleyGate101 vector, and *MdHYL1* was cloned into the pGWB406 vector. The resulting constructs were transformed into *Agrobacterium* strain C58C1 and co-infiltrated with *35S:p19* into 4-week-old *N. benthamiana* leaves. The infiltrated tobacco (*N. benthamiana*) plants were grown for an additional 3 d in a growth chamber before fluorescence signal observation. To observe the nuclear signal, 4,6-diamidino-2-phenylindole dihydrochloride was injected into the tobacco leaves 3 d after co-infiltration.

For the co-localization analysis of *MdARF17* and *MdHYL1*, *MdARF17* was inserted into pGWB455-RFP vector. The constructs (*MdARF17*-pGWB455 and *MdHYL1*-pGWB406) were used to co-infiltrate the tobacco leaves, and the co-localization signals were observed using a confocal microscope (Nikon, Tokyo, Japan). An argon laser excitation of 488 nm (intensity 1%) was examined at the wavelength of 498–540 nm to detect GFP signal, and 552 nm (intensity 1%) was examined at the wavelength of 582–647 nm to detect RFP signal.

The primers are listed in Supplemental Table S3.

In situ hybridization

In situ hybridizations of Mdm-miR160 in apple were conducted as described (Martin et al., 2009; Geng et al., 2018) with minor modifications. Briefly, the fresh leaves, stems, and roots of tissue-cultured GL-3 plants were fixed in the formalin-aceto-alcohol (FAA) fixative solution at 4°C. Decolorization was carried out with gradient alcohol and t-butanol, and embedding was performed with paraffin. The wax blocks were sliced with a slicer of 7 µm and baked at 37°C for overnight. The baked glass slides were placed in xylene to remove all the wax and then in graded series of ethanol and DEPC water to dehydrate. Finally, the slides were digested in 1 µg/mL of protease K at 37°C. For hybrid steps, glass slides were soaked in DIG-EASY-BYB buffer solution at 37°C for 30 min, then incubated with the probe at 42°C for ∼18 h, and rinsed with washing buffer at room temperature and 65°C for twice before balanced in detecting buffer and stored in TE buffer. All the reagents and buffer were from DIG Nucleic Acid Detection Kit (Roche, Basel, Switzerland) and DIG RNA Label Kit (SP6/T7).

To perform in situ hybridization of Mdm-miR160 in the micro-grafted plants, transverse sections were prepared with the stems containing the graft union, 1 cm above the union, and 1 cm below the union. In situ hybridization was performed as mentioned above.

The probes are listed in Supplemental Table S3.

RNA-seq and RT-qPCR analysis

Plant leaves obtained from GL-3 and transgenic plants under control and drought conditions were used for RNA extraction as described previously (Xie et al., 2018). RNAs were subjected to sequencing using the Illumina HiSeq platform by Novogene (Beijing, China) and aligned with the GDDH13 reference genome version 1.1 by Hisat2 (Kim et al., 2015). Differential gene expressions were analyzed by DESeq2 (Love et al., 2014) with threshold of *P*-value <0.05 and the absolute value of log2 (Fold change) >1. Fragments Per Kilobase of transcript per Million fragments mapped value was calculated by Cuffnorm in Cufflinks. Heatmaps were created by Tbtools (Chen et al., 2020). Three biological replicates were used, and each replicate has three plants.

RT-qPCR analysis was carried out as described (Xie et al., 2018). Stem–Loop RT-qPCR analysis was performed according to the previous description (Niu et al., 2019). Three biological replicates were used for each experiment.

The primers are listed in Supplemental Table S3.

5′-RLM-RACE

5′-RLM-RACE was carried out according to the manual of FirstChoice RLM-RACE Kit (Ambion, Austin, TX, USA). Briefly, RNA was extracted from the leaves of GL-3 by CTAB and then the adapter-ligated cDNA was synthesized. The second PCR products were obtained by nested PCR and purified by gel electrophoresis, and then cloned into pMD-19T vector (Takara, Shiga, Japan) for sequencing analysis (Liu et al., 2018a, 2018b).

Statistical analysis

Data are reported as the mean ± standard deviation. Statistical significance was determined by one-way ANOVA (Tukey's test) analysis using SPSS (version 21.0, USA). Variations were considered significant if $P<0.05$, 0.01, or 0.001.

Data availability

The RNA-seq data have been deposited to the NCBI with the dataset identifier PRJNA627355 and PRJNA645515.

Accession numbers

Sequence data from this article can be found in the GenBank/EMBL data libraries under accession numbers XP_008338805.2 (MdARF17), XP_008347772.2 (MdARF17-like), XP_028950144.1 (MdHYL1), XP_008385324.2 (MdARF18), XP_008359296.2 (MdARF18-like), NP_001280858.1 (MdARF3), XP_008366806.3 (MdWRKY70), XP_008383508.2 (MdWRKY71), XP_008357089.2 (MdEXP4), XP_017189299.2 (MdSPBB9), XP_008357781.1 (MdTIP1-3 like), NP_001280768.1 (MdACS), XP_028957162.1 (MdLTP3), XP_008367320.2 (MdHSP70), XP_008363740.1 (MdMADS), XP_008385625.2 (MdLAO), XP_008375333.2 (MdWAT1), XP_008390497.1 (MdATG8i), and XP_008343633.2 (MdPYL4-like).

Supplemental data

The following materials are available in the online version of this article.

Supplemental Figure S1. Protein sequence similarity of MdARF17 and MdARF17-like, expression of *MdARF17-like*, and subcellular localization of MdARF17-like.

Supplemental Figure S2. Expression of *MdARF17*, *MdARF17-like*, *MdARF18*, *MdARF18-like*, and *MdARF3* in *MdARF17* RNAi plants.

Supplemental Figure S3. Generation of *MdmARF17* OE plants.

Supplemental Figure S4. Regulation of adventitious root development by *MdARF17*.

Supplemental Figure S5. Dry weight of root and root-to-shoot ratio of *MdARF17* transgenic plants after prolonged drought treatment.

Supplemental Figure S6. Expression of Mdm-miR160a, 160b/d, and 160c in response to drought.

Supplemental Figure S7. Generation of Mdm-miR160e OE plants.

Supplemental Figure S8. Regulation of adventitious root development by Mdm-miR160.

Supplemental Figure S9. Transcriptional activities of MdARF17 and MdHYL1.

Supplemental Figure S10. Co-localization of MdARF17 and MdHYL1.

Supplemental Figure S11. Promoter analysis of *MdHYL1*.

Supplemental Figure S12. Response of *MdHYL1* to drought stress, and generation of *MdHYL1* OE and RNAi plants.

Supplemental Figure S13. Adventitious root development of *MdHYL1* transgenic plants.

Supplemental Figure S14. Dry weight of root and root-to-shoot ratio of *MdHYL1* transgenic plants after prolonged drought treatment.

Supplemental Figure S15. Plant height of *MdHYL1* transgenic plants.

Supplemental Figure S16. Leaf morphology, leaf area, leaf length, leaf width, and leaf length-width ratio of *MdHYL1* RNAi plants.

Supplemental Figure S17. Leaf morphology, leaf area, leaf length, leaf width, and leaf length-width ratio of *MdHYL1* OE plants.

Supplemental Figure S18. Transcripts of drought-responsive miRNAs in *MdHYL1* RNAi plants.

Supplemental Figure S19. Expression of *MdATG8i*, *MdPYL4-like*, *MdWRKY70*, and *MdWRKY71* in the GL-3 and *MdHYL1* RNAi plants under control and drought conditions.

Supplemental Figure S20. Expression level of *MdLAO*, *MdWAT1*, and *MdEXP4* in *MdHYL1* RNAi and *MdARF17* RNAi plants under control and drought conditions.

Supplemental Figure S21. Plant height, leaf morphology, leaf area, leaf length, leaf width, and leaf length-width ratio of GL-3 and *MdARF17* RNAi plants.

Supplemental Figure S22. Plant height, leaf morphology, leaf area, leaf length, leaf width, and leaf length-width ratio of GL-3 and *MdmARF17* OE plants.

Supplemental Figure S23. Abundance of Mdm-miRNAs in GL-3 and *MdmARF17* OE plants in response to drought.

Supplemental Figure S24. Transcripts of *MdHYL1*, Mdm-miRNAs in GL-3 and *MdARF17* RNAi *MdHYL1* RNAi plants.

Supplemental Figure S25. Expression of *MdARF17* and *MdHYL1* in transgenic calli of *MdmARF17* OE, *MdmARF17*

OE/*MdHYL1* RNAi, *MdARF17* RNAi, and *MdARF17* RNAi/*MdHYL1* RNAi.

Supplemental Figure S26. Drought tolerance of pGWB455/GL-3 and *MdHYL1* RNAi/*MdARF17* RNAi plants.

Supplemental Figure S27. Mdm-miR160 transcripts in grafted plants.

Supplemental Figure S28. Sanger sequencing peaks of Nta-miR160d.

Supplemental Figure S29. A working model for a positive feedback regulatory loop.

Supplemental Dataset S1. Upregulated genes in *MdARF17* RNAi plants under control conditions.

Supplemental Dataset S2. Downregulated genes in *MdARF17* RNAi under control conditions.

Supplemental Dataset S3. Upregulated genes in *MdARF17* RNAi under drought conditions.

Supplemental Dataset S4. Downregulated genes in *MdARF17* RNAi under drought conditions.

Supplemental Dataset S5. Upregulated genes in *MdHYL1* RNAi under drought conditions.

Supplemental Dataset S6. Downregulated genes in *MdHYL1* RNAi under drought conditions.

Supplemental Table S1. Genes upregulated in *MdARF17* RNAi plants and downregulated in *MdHYL1* RNAi plants under drought conditions.

Supplemental Table S2. Genes downregulated in *MdARF17* RNAi plants and upregulated in *MdHYL1* RNAi plants under drought conditions.

Supplemental Table S3. Primers used in this study.

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